

**IN THE UNITED STATES DISTRICT COURT
FOR THE DISTRICT OF DELAWARE**

QORVO, INC.,	)	
	)	
Plaintiff,	)	
	)	
v.	)	C.A. No. 21-1417(JPM)
	)	
AKOUSTIS TECHNOLOGIES, INC. and	)	DEMAND FOR JURY TRIAL
AKOUSTIS, INC.,	)	
	)	
Defendants.	)	
	)	

SECOND AMENDED COMPLAINT

Pursuant to Federal Rule of Civil Procedure 15(a)(2) and this Court’s Order granting Plaintiff, Qorvo, Inc. (“Plaintiff” or “Qorvo”)’s unopposed motion for leave, Qorvo, by and through its attorneys, submits this Second Amended Complaint and brings this action for damages, injunctive relief and declaratory relief against Akoustis Technologies, Inc. and Akoustis, Inc. (collectively, “Defendants” or “Akoustis”) and alleges as follows:

I. NATURE OF THE ACTION

1. This is an action for patent infringement, false advertising, false patent marking, unfair competition, trade secret misappropriation, racketeering, and civil conspiracy, respectively arising under the patent laws of the United States, 35 U.S.C. § 1, *et seq.*, the Lanham Act, 15 U.S.C. § 1125, the Defense of Trade Secrets Act (“DTSA”) (18 U.S.C. § 1832, *et seq.*), the Racketeer Influenced and Corrupt Organizations Act (“RICO”) (18 U.S.C. § 1961, *et seq.*), the North Carolina Trade Secrets Protection Act (N.C. Gen. Stat § 66-152, *et seq.*), and civil conspiracy laws of the state of North Carolina.

II. PARTIES

2. Plaintiff is a Delaware corporation with its principal place of business at 7628 Thorndike Road, Greensboro, North Carolina 27409.

3. Defendant Akoustis Technologies, Inc. is a Delaware corporation with its principal place of business at 9805 Northcross Center Court, Suite A, Huntersville, North Carolina, 28078. The Delaware registered agent for Akoustis Technologies, Inc. is Corporation Service Company, 251 Little Falls Drive, Wilmington, DE 19808.

4. Defendant Akoustis, Inc. is a Delaware corporation with its principal place of business located at 9805 Northcross Center Court, Suite A, Huntersville, North Carolina, 28078. The Delaware registered agent for Akoustis, Inc. is Corporation Service Company, 251 Little Falls Drive, Wilmington, DE 19808.

III. JURISDICTION AND VENUE

5. This Court has subject matter jurisdiction over Qorvo's patent infringement and false marking claims pursuant to 28 U.S.C. §§ 1331 and 1338(a), because they arise under the laws of the United States, specifically those relating to the infringement of U.S. patents, 35 U.S.C. § 1, *et seq.*

6. This Court has subject matter jurisdiction over Qorvo's Lanham Act claims pursuant to 28 U.S.C. § 1331, because they arise under the laws of the United States, specifically those related to the Lanham Act, 15 U.S.C. §§ 1051, *et seq.*

7. This Court has subject matter jurisdiction over Qorvo's DTSA claims pursuant to 28 U.S.C. § 1331, because they arise under the laws of the United States, specifically those related to the DTSA, 18 U.S.C. § 1832, *et seq.*

8. This Court has subject matter jurisdiction over Qorvo's RICO claims pursuant to 28 U.S.C. § 1964 and 28 U.S.C. § 1331, because they arise under the laws of the United States, specifically those related to the RICO, 18 U.S.C. § 1961 *et seq.*

9. This Court also has supplemental jurisdiction over Qorvo's state law claims pursuant to 28 U.S.C. § 1367, because the supplemental state law claims arise out of the same case or controversy as the federal claims over which this Court has original jurisdiction.

10. Akoustis is subject to this Court's personal jurisdiction, at least because Akoustis is incorporated, organized, and existing under the laws of the State of Delaware and has a registered agent in Wilmington, Delaware.

11. Venue is proper in this judicial district pursuant to 18 U.S.C. § 1965, 28 U.S.C. § 1391 and 28 U.S.C. § 1400(b) at least because Akoustis resides in this district.

IV. FACTUAL BACKGROUND

A. Qorvo

12. Qorvo is a leader in the development and commercialization of technologies and products for wireless and wired connectivity. Qorvo combines a broad portfolio of innovative radio frequency ("RF") solutions, highly differentiated semiconductor technologies, systems-level expertise and global manufacturing to supply to a diverse set of customers a broad range of products that enable a more connected world.

13. Qorvo was formed on January 1, 2015 as a result of the merger between TriQuint Semiconductor ("TriQuint") and RF Micro Devices ("RFMD"), which were formed in 1985 and 1991, respectively. Qorvo is headquartered in Greensboro, North Carolina.

14. Qorvo operates in a highly competitive industry characterized by rapid advances in technology and new product introductions. Qorvo's continued success therefore depends in substantial part on its constant attention to research and development and the creation of new

technologies. Each year, Qorvo invests significant resources on research and development activities to develop the most innovative and highest quality products in the markets in which it operates, and to establish the most efficient, repeatable, and highest quality manufacturing processes to support those products.

15. Qorvo's success depends in part on its ability to improve its products and processes faster than its competitors, anticipate changing customer requirements, and successfully develop and launch new products while reducing costs.

16. As part of its research and development, Qorvo has produced, among other cutting-edge technologies, a wide range of novel technologies and processes directed to thin-film bulk acoustic wave (BAW) resonator filters.

17. BAW resonator filters, or BAW filters, are a vital component in advanced radio frequency filtering solutions for wireless devices, as well as the world's most advanced radar and communications systems. This is because wireless devices communicate in particularized frequency bands in the electromagnetic spectrum that are assigned by government entities to prevent interference between different devices and service providers. Thus, because devices using nearby frequencies could create interference, it is important to filter out signals, through BAW filters, from those frequencies that are irrelevant to a particular device (i.e., communications in frequency bands above or below that assigned to the device) to improve communications efficacy.

18. The rapid growth of wireless devices has created an increasing demand for bandwidth, resulting in different devices using immediately adjacent bands. Thus, filter effectiveness has become a critical requirement. Moreover, prior filter designs were relatively less effective for removing unwanted frequencies. Thus, demand for more effective filters has greatly increased.

19. Qorvo's current BAW filter offerings are the result of many years of intensive, dedicated, and expensive research, design, development, testing, and refinement.

20. Qorvo's BAW filters are industry leading. Qorvo's BAW filters have been instrumental in awards received by Qorvo, including the 2015 Corporate Innovation Award (for its development of RF solutions, including BAW technology), the 2020 World Electronic Achievement Award (for Qorvo's QM28014 antenplexer, which utilizes Qorvo BAW filters), and the 2020 GTI Innovative Breakthrough Award (for Qorvo's RF Fusion™ 5G chipset, which utilizes Qorvo BAW filters). Qorvo's substantial research and development investment was (and remains) key to gaining its current position in the market and BAW filters enjoy competitive performance advantages because of such investment.

21. Qorvo protects its investments in its technology and products, including its BAW filter products, by, among other things, maintaining information regarding its products as confidential and proprietary information and filing patent applications.

22. Examples of Qorvo's confidential and proprietary information include, but are not limited to, technical expertise and know-how, such as BAW filter designs, specifications, development methods and techniques, material specifications, design tolerances, packaging, quality assurance and testing, manufacturing specifications, methods, and techniques; and business information, such as future product development and refinement plans, pricing information, cost information, marketing and sales strategies, internal organization structures, employee information, and capabilities, as well as supplier lists, customer lists, contracts, demands, desires, and requirements related to the BAW filters (together referred to herein as the "Qorvo BAW Proprietary Information").

23. Qorvo takes a number of commercially reasonable steps to maintain the confidentiality of documents containing Qorvo BAW Proprietary Information. Exemplary steps include, but are not limited to: restricting access to confidential documents within its document storage and management systems, marking its confidential documents as confidential, causing its employees and other recipients of its confidential information to enter into agreements having non-disclosure obligations, and reminding employees of their obligation to return Qorvo's confidential materials and to otherwise keep secret confidential information of Qorvo's that may be retained in their unaided memory.

24. Qorvo has also implemented device-based password protection, workspace-based password protections, role-based password protections, folder-based password protections, file-based password protections, and/or using IT systems to detect misuse and/or theft of confidential information. Where appropriate, Qorvo takes steps to limit electronic access to certain confidential/proprietary documents and information to only authorized need-to-know Qorvo employees.

25. Additionally, Qorvo's regular practice is to mark as "confidential" and/or "proprietary" any written materials containing Qorvo BAW Proprietary Information and/or other confidential information. The typical legend Qorvo uses to mark its documents recites "Qorvo™ Confidential & Proprietary Information." Qorvo's efforts to use this legend (or similar markings) are intended to serve as both a reminder to employees (or other recipients) that the materials are highly sensitive and subject to disclosure restrictions, and also to put unintended recipients and other third parties on notice that the content within the written materials are owned by Qorvo and should not be disseminated without authorization.

26. Additionally, Qorvo also provides its employees with a handbook detailing Qorvo's policy mandating that employees maintain the confidentiality of Qorvo BAW Proprietary Information and other confidential information. Further, among other steps, Qorvo enters into non-disclosure and other agreements with its employees obligating such employees not to disclose or use Qorvo BAW Proprietary Information or other confidential information without authorization and/or without making such disclosure subject to a non-disclosure or other similar agreement. Qorvo also enters into agreements with its employees that obligate them not to disclose or use Qorvo's confidential information after such time as the employees' employment with Qorvo is terminated. Upon an employee's termination, Qorvo also requests that the employee execute a document acknowledging the employee's agreement not to disclose Qorvo confidential information. Qorvo also enters into non-disclosure agreements or similar agreements with third parties (e.g., customers, suppliers, etc.) to impose non-disclosure obligations on the third parties as to Qorvo BAW Proprietary Information and other confidential information.

27. Qorvo has invested significant time, money and human capital in developing, maintaining, and protecting the Qorvo BAW Proprietary Information and other confidential information. The Qorvo BAW Proprietary Information and other confidential information described herein is strategic, proprietary information that is not generally known to the public and would not be ascertainable without the expenditure of substantial time, effort, and resources.

28. The Qorvo BAW Proprietary Information and other confidential information described herein has significant independent economic value, potential and actual, as a result of not being generally known or readily ascertainable, including by those in the industry. Qorvo's confidential information, including but not limited to the Qorvo BAW Proprietary Information and other confidential information described herein, gives Qorvo an advantage in attracting, recruiting,

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and retaining talented employees that contribute to various aspects of Qorvo's business, including but not limited to the design, testing, production, and quality of its BAW filters and other products; and it gives Qorvo an advantage over competitors in more quickly, efficiently, and effectively designing, developing, modeling, simulating, testing, qualifying, manufacturing, marketing, selling, and supporting high quality BAW filters and other products. Qorvo protects its technology and products, including its BAW filter and other products, by, among other things, maintaining information regarding its products as confidential and proprietary.

29. Qorvo's success depends in part on its ability to develop and improve its products and processes faster than its competitors, anticipate changing customer requirements, and successfully develop and launch new products while reducing cost. Akoustis' acquisition of, access to, disclosure and use of Qorvo's confidential information undermines, among other things, Qorvo's ability to do keep its competitive edge by being first to market (relative to its competitors, including Akoustis) with the highest quality filters and other products meeting customers' needs.

30. Because sales in the BAW filter market are driven, in part, by how quickly the supplier can develop the filter meeting customers' ever increasing requirements, Qorvo BAW Proprietary Information, would be highly valuable to a startup company trying to enter the market, such as Akoustis. Indeed, acquiring Qorvo BAW Proprietary Information would enable a startup company to skip over many of the laborious, time-exhaustive and highly expensive trial and error processes required to design BAW filters to satisfy such requirements—processes that every other company has to endure to fairly compete. And while Qorvo BAW Proprietary Information would be highly valuable to any BAW filter market entrant, it is of particularly high value to a new market entrant that intends to offer products that directly compete with and provide a substitute for the precise BAW filters that Qorvo sells, and that were developed on the basis of the Qorvo BAW

Proprietary Information. Effectively, a startup could use Qorvo BAW Proprietary Information to unfairly advance its ability to compete earlier and more aggressively with market participants—and, as described below, Akoustis has done exactly that.

31. To protect its valuable intellectual property, Qorvo also often seeks patent protection. Qorvo has applied for and has been awarded over 3,000 patents in the United States and abroad through its history (including through TriQuint and RFMD), and including hundreds of patents related to its BAW filter technology.

32. Two of these patents, U.S. Patent Nos. 7,522,018 (the “’018 Patent”) and 9,735,755 (the “’755 Patent”) describe important structural features of BAW filters that enable the performance required by today’s wireless devices. A copy of each patent (collectively, the “Patents-in-Suit”) is attached hereto as Exhibits A (the ’018 Patent) and Exhibit B (the ’755 Patent).

33. The ’018 Patent (Exhibit A), entitled “Electro-Acoustic Resonator With A Top Electrode Layer Thinner Than A Bottom Electrode Layer,” was duly and legally issued on April 21, 2009 from an application filed on December 4, 2003, naming Robert Milsom and Hans-Peter Löbl as the inventors.

34. The ’018 Patent claims priority to European Application No. 02258613, filed December 13, 2002.

35. Qorvo owns, by assignment, all substantial right, title, and interest in and to the ’018 Patent.

36. Pursuant to 35 U.S.C. § 282, the ’018 Patent is presumed valid.

37. Claim 1 of the ’018 Patent reads as follows:

1. Electro-acoustic resonator (1, 8, 17) comprising a membrane structure FBAR (1) with a layer structure comprising a piezoelectric layer (5, 14, 24)

and a top (6, 15, 25) and a bottom (4, 13, 23) electrode layer, with the thickness (T1, T2, . . . T6) of the two electrode layers being unequal, characterised in that the top electrode layer (T1, T3, T5) is thinner than the bottom (T2, T4, T6) electrode layer to increase a filter bandwidth of the electro-acoustic resonator.

38. The '755 Patent (Exhibit B), entitled "BAW Resonator Having Lateral Energy Confinement And Methods of Fabrication Thereof," was duly and legally issued on August 15, 2017 from an application filed on October 6, 2015, naming Gernot Fattinger and Alireza Tajic as the inventors.

39. The '755 Patent claims priority to provisional application No. 62/207,702, filed August 20, 2015.

40. Qorvo owns, by assignment, all substantial right, title, and interest in and to the '755 Patent.

41. Pursuant to 35 U.S.C. § 282, the '755 Patent is presumed valid.

42. Claim 9 of the '755 Patent reads as follows:

9. A BAW resonator comprising:
a piezoelectric layer;
a first electrode on a first surface of the piezoelectric layer;
a second electrode on a second surface of the piezoelectric layer opposite the first electrode on the first surface of the piezoelectric layer;
a passivation layer on a surface of the second electrode opposite the piezoelectric layer within an active region of the BAW resonator, the passivation layer having a thickness (T_{PA}) within the active region of the BAW resonator; and
one or more material layers on the second surface of the piezoelectric layer adjacent to the second electrode in an outer region of the BAW resonator, the outer region of the BAW resonator being a region outside of the active

region of the BAW resonator and the one or more material layers having a thickness that is n times the thickness (T_{PA}) of the passivation layer within the active region, wherein:

n is a value other than 1; and

n is such that the outer region of the BAW resonator and the active region of the BAW resonator are acoustically matched in such a manner that one or more wavelengths that cause energy leakage into the outer region are not excited in the active region.

43. Qorvo's considerable investment in the development, manufacturing, and marketing of its BAW filters, as well as its efforts to maintain the confidentiality of the Qorvo BAW Proprietary Information and patent other innovations related to its BAW filters—such as through the Patents-in-Suit—has resulted in substantial sales for Qorvo in the BAW filter market.

44. Qorvo's business is also dependent on its ability to attract and keep key technical personnel and management. Qorvo has invested substantial time, effort, and expense in developing and training its employees' knowledge and experience in the areas crucial to Qorvo's BAW filter technologies, including in RF engineering, integrated circuit and filter design, and technical marketing and support.

B. Akoustis

45. Defendant Akoustis was founded in 2014 by Jeffrey B. Shealy, Steven P. Denbaars, and Richard T. Ogawa.

46. Mr. Shealy was, from October 2001 through February 2014, a vice-president of RFMD which, together with TriQuint, merged to become Qorvo in 2015. By virtue of his leadership role at RFMD, Mr. Shealy had extensive knowledge of RFMD's and Qorvo's operations, business strategies, patents, and personnel as of the date of his departure from Qorvo to form Akoustis.

47. Mr. Shealy also sits on Akoustis' Board of Directors, a majority of which are Ex-Qorvo employees. Beyond Mr. Shealy, Co-Chairman Jerry D. Neal founded RFMD in 1991 and served in various leadership roles, including as Executive Vice President of Marketing and Strategic Development at the time of his departure. Board member Arthur E. Geiss was Vice President of Wafer Fab Operations at RFMD until his departure. Board member Suzanne Rudy was Vice President of Tax and Corporate Treasurer, Compliance Officer, and Assistant Secretary at Qorvo at the time of her departure. Mr. Neal, Mr. Geiss, and Ms. Rudy had extensive knowledge of Qorvo's (including RFMD's) operations, business strategies, patents, and personnel as of their dates of departure.

48. Akoustis was formed specifically to compete with Qorvo (and its predecessor companies) in the BAW filter market. While Akoustis describes itself as being focused on "developing, designing, and manufacturing ... RF filter products for the wireless industry," its current product offerings are narrowly limited to what it calls its "3-7 GHz BAW RF filter portfolio."

49. Akoustis' "3-7 GHz BAW RF filter portfolio" currently includes twelve BAW filters. See ALL PRODUCTS - AKOUSTIS TECHNOLOGIES, <https://akoustis.com/products/filters/applications/all-products/> (last visited Sept. 14, 2021).

50. Akoustis' BAW filters compete with Qorvo's BAW filters.

51. While Qorvo's design and development of BAW filters has taken many years and great expense, Akoustis was able to design and qualify its 3-7 GHz BAW RF filter portfolio from scratch in just a few years and with just a fraction of the number of Qorvo's employees and a fraction of the amount of Qorvo's research and development investment. To do so, Akoustis

improperly leveraged Qorvo's intellectual property, including the Patents-in-Suit and the Qorvo BAW Proprietary Information, as obtained by its systematic poaching of Qorvo's employees.

52. Akoustis built its 3-7 GHz BAW RF filter portfolio using Qorvo's patented technology. Each of Akoustis' BAW filters identified above infringes the Patents-in-Suit. Each of Akoustis' BAW filters uses the structural configurations disclosed and claimed by the Patents-in-Suit to provide the performance necessary to compete in the BAW filter market. On information and belief, Akoustis has been aware of Qorvo's patent portfolio throughout its existence, including specifically the Patents-in-Suit.

53. On information and belief, Akoustis has also used the Qorvo BAW Proprietary Information to design, manufacture, market, and sell its BAW filters and other offerings, and to otherwise develop its technological know-how.

C. Akoustis' Poaching of Qorvo Employees

54. Since its inception, Akoustis has engaged in conscious, methodical, and systematic recruitment and solicitation of Qorvo's employees in all major areas. Indeed, and beyond a majority of its board being Ex-Qorvo employees, Akoustis has poached over twenty key employees, including crucial personnel from engineering, operations, sales, quality control and testing, packaging, and management.

55. On information and belief, Akoustis recruited Ex-Qorvo employees for positions with largely overlapping responsibilities to their positions at Qorvo, focusing on employees with the specialized knowledge of Qorvo's confidential and proprietary information, including the Qorvo BAW Proprietary Information, that Akoustis needed at the particular stage of its development. For example, Qorvo focused on those Qorvo employees with business development and BAW filter design experience when it was initially defining and designing its products, and

then focused on those Qorvo employees with quality, manufacturing, and management expertise as it moved from product design to manufacture and sales.

56. As examples, since its formation, Akoustis has poached Qorvo employees (the “Ex-Qorvo BAW Employees”) with the following job titles/descriptions.

	Job Title	Job Description
1	Director, Business Development	developed, coordinated, and implemented plans designed to increase existing business and capture new opportunities
2	Device Engineer	evaluated and characterized high power and mm-wave GaN RF technologies; supported foundry customers in evaluation and selection of RFMD GaAs and GaN semiconductor processes; planned, implemented, and managed GaN device performance and reliability projects; generated production test plans for customer discrete FET and MMIC designs; developed and supported customer qualification tests
3	Plasma Process Engineer	analyzed BAW resonator electrical data and metrology data; developed new processes on plasma platforms; improved process reliability; demonstrated proficiency in design of experiment, statistical process control, and model based problem solving
4	Customer Quality Manager	managed customer concerns and ensured their satisfaction; traveled extensively to train local teams and to work with customers; lead team responsible for chemical compliance monitoring and reporting
5	Senior Tax Manager	managed all federal, foreign, and state & local income tax filings; managed audit activities; responsible for corporate cash planning, investment, and funds management; managed all sales and use and property tax functions for the US; and responsible for expatriate tax compliance
6	Senior Business Analyst, SAP IBP	managed implementation of SAP integrated business planning; supported planning team regarding demand and supply; assisted with revenue planning and margin analysis
7	Director of Infrastructure and Defense Packaging	evaluated current and future enterprise infrastructure needs
8	Senior BAW Design Engineer	responsible for the research and design of new BAW products and systems
9	Senior Planner	responsible for the execution of in-house and external prototype assembly

	Job Title	Job Description
10	Senior RF Test Engineer	responsible for hardware and software development for 2G/3G/4G power amplifiers; developed production test solutions; performed production line debug and efficiency optimization; performed product data analysis; and performed production
11	Senior Customer Quality Engineer	acted as the representative to premier customers; provided strategic direction and leadership; and knowledgeable in quality, design, supply chain, product engineering, sales, and marketing
12	Manager, BAW R&D, Technology	managed technology divisions and BAW research and development

57. Each of the Ex-Qorvo BAW Employees identified above had routine and repeated access to one or more categories of the Qorvo BAW Proprietary Information.

58. On information and belief, numerous Ex-Qorvo BAW Employees who are now employed by Akoustis were aware of the Patents-in-Suit during their work at Qorvo. For example, the Ex-Qorvo BAW Employees include employees who formerly worked in the same department at Qorvo as the inventors of the Patents-in-Suit and those Ex-Qorvo BAW Employees now work in a similar department at Akoustis that is involved in developing the technology embodied in the Accused Products. As part of their former employment with Qorvo, these Ex-Qorvo BAW Employees would have been aware of the Patents-in-Suit. This knowledge then transferred to Akoustis when they became Akoustis employees.

59. Additionally, on information and belief, Akoustis has had knowledge of the Patents-in-Suit because a number of senior executives of Akoustis were executives at Qorvo (or one of its predecessor companies, such as RFMD) at the time when the Patents-in-Suit were issued or acquired. On information and belief, these former Qorvo executives had knowledge of the Patents-in-Suit in connection with their positions at Qorvo and brought that knowledge to Akoustis.

60. The Ex-Qorvo BAW Employees entered into non-disclosure agreements as a condition of their employment. The non-disclosure agreements require the Ex-Qorvo BAW Employees not to use, take, or disclose Qorvo's confidential information, including the Qorvo BAW Proprietary Information, without authorization.

D. Akoustis Engages in Widespread Misappropriation of Qorvo's Trade Secrets

61. Akoustis has built its company based on the misappropriation of Qorvo BAW Proprietary Information and other confidential information, using that information to shortcut the hard work of building an innovative company and to shorten the typical time to market for new products in the relevant market.

62. Discovery has confirmed the blatant and widespread nature of Akoustis' efforts to steal Qorvo's trade secrets and its actual use of the same throughout the company's operation. Indeed, this disregard for the propriety of Qorvo's trade secrets has been found in nearly every area of Akoustis' operations, with direct evidence of Qorvo's trade secrets being misappropriated at the highest of levels within the company, including but not limited the CEO of Akoustis, Jeffrey Shealy and other members of the executive team. Akoustis' files are littered with documents that are expressly marked as Qorvo's "Confidential" and "Proprietary" information. Akoustis employees unabashedly acknowledge in internal communications that certain documents are Qorvo's confidential files. Akoustis employees nonetheless used the confidential information. Akoustis employees even re-purposed a highly confidential Qorvo document by stripping-off the Qorvo confidentiality legend and inserting Akoustis' own logo on the Qorvo document. As set forth below, Qorvo has identified at least 86 trade secrets that Akoustis has misappropriated, and Qorvo believes that recent and ongoing discovery will show Akoustis' misappropriation of additional trade secrets.

63. One of the primary means Akoustis used to misappropriate Qorvo BAW Proprietary Information and other confidential information was to strategically hire key Qorvo employees. For example, Akoustis has poached the Ex-Qorvo BAW Employees for employment in positions with substantial responsibility overlap to those employees' previous roles at Qorvo. On information and belief, the Ex-Qorvo BAW Employees were hired for the specific purpose of using their knowledge of the Qorvo BAW Proprietary Information to permit Akoustis to compete directly against Qorvo for customers and market share.

64. Akoustis has aggressively recruited Qorvo employees to obtain confidential information. On several occasions, Akoustis proactively contacted employees with access to Qorvo BAW Proprietary Information—sometimes repeatedly—as part of this recruitment scheme. For example, in late 2020, an Akoustis engineering manager contacted a Qorvo Product Engineer through LinkedIn. During a subsequent call, the Akoustis engineering manager expressly asked the Qorvo Product Engineer to disclose Qorvo BAW Proprietary Information. More specifically, the Akoustis engineering manager asked this Qorvo employee to access the corporate yield page on Qorvo's computer system and provide screen shots of information found on that page. The corporate yield page is a repository for confidential information. It reflects, for example, analytical tools Qorvo is using to assess its products, the parts that Qorvo is using, the tests that Qorvo is running on those parts, and the performance and failure rates of those parts. There is no legitimate reason that Akoustis would ask a Qorvo employee to provide this type of confidential information as part of the recruitment process. The Qorvo Product Engineer felt that Akoustis' request for the confidential information was unethical and reported the incident to a supervisor.

65. The sheer number of highly-knowledgeable and qualified Qorvo employees poached by Akoustis over its history has resulted in Akoustis inevitably using their knowledge and

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experience, including the Qorvo BAW Proprietary Information. It is highly probable—and on information and belief, inevitable—that Akoustis has used, relied upon, and exploited the Ex-Qorvo BAW Employees’ knowledge of the Qorvo BAW Proprietary Information. In addition to their knowledge of and experience with Qorvo BAW Proprietary Information, these employees had been entrusted with access to documents containing Qorvo BAW Proprietary Information and other confidential information. Upon information and belief, during the time period prior to their departure from Qorvo and their joining Akoustis, these employees had a specific opportunity to improperly access and take Qorvo documents containing Qorvo BAW Proprietary Information and other confidential information, and the trade secrets within them, including for disclosure or use at Akoustis without the express or implied consent or authority of Qorvo. Furthermore, at least because Akoustis recruited and hired these employees (those who had accessed and taken such Qorvo documents), Akoustis, as their new employer, had a specific opportunity to improperly obtain and use the documents, and the trade secrets within them, without the express or implied consent or authority of Qorvo.

66. For example, in October 2019, Akoustis recruited Robert Dry. Mr. Dry was the Director of Qorvo’s Infrastructure and Defense Packaging business. In that position, Mr. Dry had access and was aware of Qorvo BAW Proprietary Information, including but not limited to micro BAW processes used to shrink the size of filters. Mr. Dry also had access to confidential Qorvo documents reflecting Qorvo’s trade secrets, including but not limited to Qorvo BAW Proprietary Information and other confidential information described herein. As Akoustis attempted to manufacture competing products, Akoustis was in need of the specific kind of confidential information known to Mr. Dry. This technology is not publicly available and is highly valuable. Mr. Dry was also a member of the patent committee at Qorvo and one of the persons who

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determined whether inventions should be maintained as trade secrets or disclosed in patent applications. As such, Mr. Dry was in a position to have widespread knowledge of Qorvo BAW Proprietary information. On information and belief, Akoustis actively recruited Mr. Dry specifically because of his knowledge of Qorvo's proprietary information and placed him in the position of Vice President of Operations with the expectation that Mr. Dry would use Qorvo BAW Proprietary information to the benefit of Akoustis, including micro BAW technology.

67. After Akoustis targeted and hired him, Mr. Dry, in breach of his obligations to Qorvo, distributed confidential Qorvo documents within Akoustis, including but not limited to documents relating to the manufacture of Qorvo's BAW filters, such as documents concerning assembly, packaging, and inspection of BAW filters and documents relating to customer requirements for such BAW filters. Upon information and belief, Mr. Dry took the Qorvo documents without authorization while he was employed by Qorvo. The confidential Qorvo documents Mr. Dry stole and shared with Akoustis were expressly marked confidential, and contained trade secret information. For at least this reason, Mr. Dry knew or should have known that the information in such documents included one or more Qorvo trade secrets and that he was not authorized to distribute the documents to Akoustis personnel. Nevertheless, Mr. Dry distributed confidential Qorvo documents in response to requests from other Akoustis employees. Upon information and belief, Akoustis then used the confidential Qorvo documents in connection with developing BAW filters that compete with Qorvo's products. As the Vice President Operations and Test at Akoustis, Mr. Dry is one of just ten members of Akoustis' "management team."

68. By way of further example, in November 2020, Akoustis recruited David Breton. Mr. Breton was a Manager of BAW Research and Development at Qorvo. In that position, Mr.

[REDACTED]

Breton had access to and was aware of Qorvo BAW Proprietary Information, including but not limited to how Qorvo filters achieve superior performance in high frequency bands (5-7 GHz). Mr. Breton was also knowledgeable about the customized, proprietary, and highly confidential tools used by the Developmental Physics Team at Qorvo to perform simulations as part of the technology development process. As Akoustis is engaged in designing, developing and manufacturing competing products in the 5-7 GHz frequency band, Akoustis was in need of the specific kind of confidential information known to Mr. Breton. The Qorvo BAW Proprietary Information known to Mr. Breton is not publicly available and is highly valuable. On information and belief, Akoustis actively and specifically recruited Mr. Breton and placed him in the position of Manager with the expectation that Mr. Breton would use Qorvo BAW Proprietary Information, including confidential information concerning how Qorvo simulates and achieves improved performance in its BAW filters, to the benefit of Akoustis.

69. By way of further example, in November 2019 Akoustis recruited Guillermo Moreno. Mr. Moreno was a Senior BAW Design Engineer at Qorvo. In that position, Mr. Moreno had access to and was aware of Qorvo BAW Proprietary Information, including but not limited to how Qorvo BAW filters achieve superior performance. The Qorvo BAW Proprietary Information known to Mr. Moreno is not publicly available and is highly valuable. As Akoustis engaged in designing, developing and manufacturing competing BAW products, Akoustis was in need of the specific kind of confidential information known to Mr. Moreno. On information and belief, Akoustis actively recruited Mr. Moreno and placed him in the position of Design Engineering Manager with the expectation that Mr. Moreno would use Qorvo BAW Proprietary information to the benefit of Akoustis.

70. By way of further example, in May 2020 Akoustis recruited William Schmid. Mr. Schmid was a Senior RF Test Engineer at Qorvo. In that position, Mr. Schmid had access to and was aware of Qorvo BAW Proprietary Information, including but not limited to the proprietary procedures and software libraries that Qorvo uses to test products. The Qorvo BAW Proprietary Information known to Mr. Schmid is not publicly available and is highly valuable. In its business cycle, Akoustis was in need for someone specifically having knowledge of Qorvo's confidential and proprietary information relating to testing of BAW filters. Accordingly, on information and belief, Akoustis actively recruited Mr. Schmid and placed him in the position of Operations Test Senior Manager with the expectation that Mr. Schmid would use Qorvo BAW Proprietary information, including Qorvo's testing procedures and information concerning Qorvo's software libraries.

71. By way of further example, in July 2020, Akoustis recruited Kindra Lane. Ms. Lane was a Senior NPI Planner at Qorvo. In that position, Ms. Lane worked on new product development and was responsible for worldwide logistics. Ms. Lane had access to and was aware of Qorvo BAW Proprietary Information, including but not limited to Qorvo's EFC modeling tool, which is a secret and confidential estimation process Qorvo has built to estimate costs for products. Ms. Lane was also aware of Qorvo's confidential internal financial targets and mid- and long-term plans, proprietary cost issues relating to the manufacture of BAW filters, and the identities of Qorvo's critical vendors. The Qorvo BAW Proprietary Information known to Ms. Lane is not publicly available and is highly valuable. Akoustis was in specific need of the Qorvo BAW Proprietary Information known to Ms. Lane, including Qorvo's proprietary modeling processes and information related to sourcing components and materials from world-wide suppliers critical to the manufacture of BAW filters. Accordingly, on information and belief, Akoustis actively and

specifically recruited Ms. Lane and placed her in the position of Operations Planning Manager with the expectation that Ms. Lane would use Qorvo BAW Proprietary information, including information concerning Qorvo's EFC modeling tool, financial targets, vendors, and mid- and long-term plans to the benefit of Akoustis.

72. By way of further example, in April 2019 Akoustis recruited Wendy Wright. Ms. Wright was a Senior Business Analyst at Qorvo. In that position, Ms. Wright was responsible for implementing and customizing Qorvo's integrated business planning ("IBP") system. Ms. Wright had access to and was aware of Qorvo BAW Proprietary Information, including but not limited to Qorvo's customized IBP system and Qorvo's proprietary process for generating "what if" scenarios to analyze demand and supply. The Qorvo BAW Proprietary Information known to Ms. Wright is not publicly available and is highly valuable. On information and belief, Akoustis actively recruited Ms. Wright and placed her in the position of Global Planning Manager with the expectation that Ms. Wright would use Qorvo BAW Proprietary information, including information concerning Qorvo's IBP system and "what if" analysis to the benefit of Akoustis.

73. By way of further example, in August 2020 Akoustis recruited Paul Makowenskyj. Mr. Makowenskyj was a Senior Customer Quality Engineer at Qorvo. In that position, Mr. Makowenskyj had access to and was aware of Qorvo BAW Proprietary Information, including but not limited to Qorvo's product pipeline, the performance and requirements for those products, and Qorvo's interactions with customers relating to quality issues of BAW filters. The BAW Proprietary Information known to Mr. Makowenskyj is not publicly available and is highly valuable. On information and belief, Akoustis was in specific need of the Qorvo BAW Proprietary Information known to Mr. Makowenskyj, and therefore actively recruited Mr. Makowenskyj and placed him in the position of Staff Quality Engineer with the expectation that Mr. Makowenskyj

would use Qorvo BAW Proprietary Information, including information concerning Qorvo's product pipeline, the performance and requirements for those products, and Qorvo's customers, for the benefit of Akoustis.

74. The former Qorvo employees identified in the preceding paragraphs are only recent examples of Akoustis' efforts to raid Qorvo employees to obtain confidential information that would allow Akoustis to short-cut its entry into the BAW filter market. In addition to the former Qorvo employees listed above, Akoustis has also hired at least the following employees away from Qorvo: Tony Espinoza, Ali-Imran Bawangaonwala, Todd Bender, Shawn Gibb, David Hodge, Rohan Houlden, Joonbum Kwon, John Myrick, and Ya Shen. On information and belief, each of these former Qorvo employees had access to Qorvo BAW Proprietary Information that would benefit Akoustis.

75. After Rohan Houlden left Qorvo to join Akoustis, and in breach of his obligations to Qorvo, he distributed Qorvo confidential information and documents within Akoustis, including but not limited to documents relating to the assembly, packaging, inspection, branding, marketing and sales of Qorvo's BAW filters, documents containing intelligence reports prepared by Qorvo that include information regarding Qorvo's competitive position with its customers, feedback from Qorvo's customers, and market intelligence and strategy, and as well as documents containing Qorvo financial information, including salary information for substantially all of Qorvo's engineering roles. By way of non-limiting example, Mr. Houlden stole and distributed within Akoustis some of Qorvo's confidential-marked test plan and product requirements documents for designing, developing, testing, and characterizing Qorvo's products, including but not limited to its BAW Filters. In addition, Mr. Houlden stole and distributed within Akoustis and to Akoustis' third-party recruiters a confidential-marked Qorvo document containing comprehensive

[REDACTED]

compensation and job structure information for all of Qorvo's engineering and sales jobs in various important markets, including the United States. Upon information and belief, Mr. Houlden obtained such information and documents, among others, while employed by Qorvo and/or via unauthorized channels after leaving Qorvo to join Akoustis. Upon information and belief, on multiple occasions, Mr. Houlden solicited Qorvo confidential information and documents from individuals with access to such information in violation of confidentiality obligations and then distributed such information and documents within Akoustis. At least in part, Mr. Houlden also had access to the foregoing Qorvo confidential information via his prior employment with Qorvo. Upon information and belief, Akoustis then used the confidential Qorvo documents in connection with developing BAW filters that compete with Qorvo's products. As the Chief Product Officer at Akoustis, Mr. Houlden was one of approximately ten members of Akoustis' "management team."

76. Akoustis' Executive Vice President of Business Development and Corporate Officer, Dave Aichele, also distributed Qorvo confidential-marked documents on multiple occasions to others at Akoustis, including to Akoustis' CEO Jeffrey Shealy. Such confidential-marked documents relate to, for example, Qorvo's product roadmaps, BAW filter design and simulations, and confidential intelligence reports prepared by Qorvo that include sensitive information regarding Qorvo's competitive position with its customers, feedback from Qorvo's customers, and market intelligence and strategy. Mr. Aichele disseminated this information within Akoustis without Qorvo's permission or authorization. As former Qorvo employees, all of these Akoustis executives and officers were privy to the sensitivity and value of the trade secret information in these confidential Qorvo documents, and knew or should have known that Akoustis was not authorized to acquire, access, disclose, disseminate, or otherwise use the same and that doing so would harm Qorvo's competitive position.

77. Akoustis has also made public statements that indicate it has obtained Qorvo BAW Proprietary Information, including confidential information about Qorvo's product pipeline and development activities. For example, in a December 2021 issue of Wireless Watch, Akoustis' Vice President of Corporate Development stated that Akoustis WiFi 6E products will outperform future Qorvo WiFi 6E products: "Sepenzis claimed that Qorvo's BAW filter can only operate with a low temperature co-fired ceramic (LTCC) filter placed on top. He described the LTTC component was 'low performance,' noting that while it can cover the 5GHz and 6GHz bands, it cannot separate them so that they can be used simultaneously, which renders the addition of an extra band somewhat useless." Mr. Sepenzis did not provide any basis for his statement and Qorvo does not have a WiFi 6E 6GHz product on the market. As such, Mr. Sepenzis's statement about potential future Qorvo products is, on information and belief, based on Qorvo BAW Proprietary Information concerning Qorvo's development pipeline and prototyping process.

78. In addition to the foregoing, in many cases Akoustis has misappropriated highly valuable trade secrets contained within sensitive Qorvo documents that are very clearly marked as Qorvo confidential and proprietary information. Upon information and belief, it obtained these documents through Ex-Qorvo employees it recruited. A sample of such Qorvo trade secrets, the documents they are reflected in, and the blatant misappropriations of the same at Akoustis are detailed below.

Trade Secrets 1-18 (Exhibits C-1, C-2, C-3)

79. Through years of research and development, and at great expense, Qorvo developed trade secrets related to the manufacture and production of semiconductor devices, including the inspection of products such as wafers used for BAW filters, and procedures for inspection and rejection analysis of the same to reduce production delays and minimize waste. These trade secrets

[REDACTED]

existed in written form at least as early as January 20, 2020, including as reflected in a confidential-marked Qorvo document attached hereto as Exhibit C-1 and bearing the bates label QORVO_00035002-069.

80. Qorvo trade secrets contained in Exhibit C-1 include, but are not limited to: (a) the overall process and sequence for inspecting wafers and handling wafers with [REDACTED] [REDACTED] (“Trade Secret 1”), *see, e.g.*, Exhibit C-1, QORVO_00035007-013 at Sections 7.1.4-7.1.6; (b) optimal steps to take in [REDACTED] [REDACTED] at various levels of inspection before shipment (“Trade Secret 2”), *see, e.g., id.*; (c) how certain inspection criteria can be satisfied for [REDACTED] (“Trade Secret 3”), *see, e.g.*, Exhibit C -1, QORVO_00035007 at Section 7.1.2(1); (d) [REDACTED] [REDACTED] of inspection criteria failures (“Trade Secret 4”), *see, e.g.*, Exhibit C-1, QORVO_00035007 at Section 7.1.3; (e) the conditions under which a wafer should be [REDACTED] [REDACTED] (“Trade Secret 5”), *see, e.g.*, Exhibit C-1, QORVO_00035007-011 at Section 7.1.4; (f) additional testing that may be required upon discovery of [REDACTED] (“Trade Secret 6”), *see, e.g., id.*; (g) conditions to satisfy before a defective wafer can be [REDACTED] (“Trade Secret 7”), *see, e.g.*, Exhibit C-1, QORVO_00035007-012 at Section 7.1.4-7.1.5; (h) which defects should be considered [REDACTED] [REDACTED] (“Trade Secret 8”); *see, e.g., id.*; (i) procedures for handling specific types of failures or defects (“Trade Secret 9”), *see, e.g.*, Exhibit C-1, QORVO_00035012-013 at Section 7.1.6; (h) optimal operating conditions for testing equipment (“Trade Secret 10”), *see, e.g.*, Exhibit C-1, QORVO_00035007-014 at Section 7.1; (i) [REDACTED] conditions, defects, and characteristics to consider when performing

[REDACTED]

wafer inspections (“Trade Secret 11”), *see, e.g.*, Exhibit C-1, QORVO_00035014-019 at Sections 7.2.1-7.2.2; (j) [REDACTED] conditions, defects, and characteristics to consider when performing wafer inspections (“Trade Secret 12”), *see, e.g.*, Exhibit C-1, QORVO_00035019-020 at Section 7.2.3; (k) [REDACTED] conditions, defects, and characteristics to consider when performing wafer inspections (“Trade Secret 13”), *see, e.g.*, Exhibit C-1, QORVO_00035020-21 at Section 7.2.4; (l) [REDACTED] conditions, defects, and characteristics to consider when performing wafer inspections (“Trade Secret 14”), *see, e.g.*, Exhibit C-1, QORVO_00035021-23 at Section 7.2.5; (m) [REDACTED] conditions, defects, and characteristics to consider when performing wafer inspections (“Trade Secret 15”), *see, e.g.*, Exhibit C-1, QORVO_00035023 at Sections 7.2.6; (n) [REDACTED] rejection details (“Trade Secret 16”), *see, e.g.*, Exhibit C-1, QORVO_00035023 at Sections 7.2.7, (o) images of real world inspection observations illustrating how to [REDACTED] (“Trade Secret 17”), *see, e.g.*, Exhibit C-1, QORVO_00035030-059; and (p) specific post-inspection documentation to maintain (“Trade Secret 18”) *see, e.g.*, Exhibit C-1, QORVO_00035024-027.

81. Qorvo has taken and continues to take commercially reasonable steps to protect Trade Secrets 1-18. As set forth in paragraphs 1 through 29, Qorvo protects its trade secrets at least in part by maintaining the confidentiality of documents containing such information.

82. As a result none of Trade Secrets 1-18 were or are generally known or readily ascertainable, nor can they be obtained through the public domain. Nor could Trade Secrets 1-18 be determined through reverse engineering. Trade Secrets 1-18 derive independent economic value from not being generally known to, and not being readily ascertainable through proper means by, entities who can obtain economic value from the disclosure or use of the information. Trade Secrets 1-18 create competitive value for Qorvo by enabling its manufacturing team to efficiently

[REDACTED]

and accurately inspect its products and determine whether such products have been manufactured to an acceptable standard. This information, in turn, informs Qorvo's engineers in making modifications and improvements to the manufacturing process to avoid the issues leading to unacceptable products. This enables Qorvo to, among other things, create more reliable and more robust products that can perform with greater resilience when deployed in real world environments, and gives Qorvo a competitive edge in the industry. In short, Trade Secrets 1-18 enable Qorvo to create products, including BAW filters, of a high level of quality that drives its business. Moreover, Trade Secrets 1-18 also create competitive value for Qorvo [REDACTED]

[REDACTED]

[REDACTED] thereby avoiding waste and saving resources to use elsewhere in the business. Trade Secrets 1-18 took Qorvo several years to develop, and cost Qorvo millions of dollars to create. By misappropriating Trade Secrets 1-18, Akoustis obtained an unfair advantage, including but not limited to obtaining a head start in its ability to produce and offer semiconductor products that realize the same efficiencies and cost savings, and to avoid failures resulting from missed quality considerations. Moreover, Akoustis avoided the great costs and expenses of developing its own inspection procedures and criteria through trial and error.

83. Akoustis misappropriated Trade Secrets 1-18 by improperly acquiring, accessing, disclosing and using such trade secrets for its own benefit. Upon information and belief, Akoustis' misappropriation is continuing. Akoustis and its employees knew or had reason to know that Trade Secrets 1-18 were acquired by improper means, including through at least one Ex-Qorvo employee who improperly disclosed Trade Secrets 1-18 with Akoustis personnel. At least because these Ex-Qorvo employees had access to Qorvo documents and trade secrets during their employment with Qorvo, such Ex-Qorvo employees had a specific opportunity to acquire the same for disclosure or

[REDACTED]

use without the express or implied consent or authority of Qorvo. Furthermore, at least because Akoustis recruited and hired these employees (those who had accessed and taken such Qorvo documents), Akoustis, as their new employer, had a specific opportunity to improperly obtain and use the documents, and the trade secrets within them, without the express or implied consent or authority of Qorvo.

84. [REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

85. [REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

86. Upon information and belief, in addition to acquiring, accessing, and disclosing Qorvo's trade secret information, Akoustis knowingly used and continues to use Qorvo's trade secret information to advance its business.

Trade Secrets 19-24 (Exhibits D-1, D-2, D-3)

87. Through years of research and development, and at great expense, Qorvo developed trade secrets related to BAW filter testing. These trade secrets existed in written form at least as early as September 25, 2015, including as reflected in a confidential-marked Qorvo document attached hereto as Exhibit D-1 and bearing the bates label QORVO_00035070-105.

88. Qorvo trade secrets contained in Exhibit D-1 include, but are not limited to: (a) architectural details of a system that was devised by Qorvo [REDACTED] [REDACTED] ("Trade Secret 19"), see Exhibit D-1 at QORVO_00035084; (b) instructions regarding how to implement and use the aforementioned system ("Trade Secret

[REDACTED]

20”), *see* Exhibit D-1 at QORVO_00035084, 086; (c) layout and set up details of RF filter power testing, [REDACTED] (“Trade Secret 21”), *see* Exhibit D-1 at QORVO_00035084-085, 079, 093; (d) instructions concerning [REDACTED] [REDACTED] such RF testing [REDACTED] [REDACTED] (“Trade Secret 22”), *see* Exhibit D-1 at QORVO_00035081, 087-088, 094; (e) [REDACTED] correlations (“Trade Secret 23”), *see* Exhibit D-1 at QORVO_00035084; and (f) RF filter data, test results, and findings resulting therefrom (“Trade Secret 24”), *see* Exhibit D-1 at QORVO_00035098-105.

89. Qorvo has taken and continues to take commercially reasonable steps to protect Trade Secrets 19-24. As set forth in paragraphs 1 through 29, Qorvo protects its trade secrets at least in part by maintaining the confidentiality of documents containing such information.

90. As a result, none of Trade Secrets 19-24 were or are generally known or readily ascertainable, nor can they be obtained through the public domain. Nor could Trade Secrets 19-24 be determined through reverse engineering. Trade Secrets 19-24 derive independent economic value from not being generally known to, and not being readily ascertainable through proper means by, entities who can obtain economic value from the disclosure or use of the information. Trade Secrets 19-24 create competitive value for Qorvo by [REDACTED]

[REDACTED]

[REDACTED] This information, in turn, informs Qorvo’s design engineers in improve their processes, identify new rejection/acceptance criteria, and to track and take preventative steps to avoid the issues resulting in a particular defect. This enables Qorvo to create more reliable and more robust BAW filters that can perform with greater resilience when deployed in real world environments, and giving Qorvo

[REDACTED]

a competitive edge in the industry. In short, Trade Secrets 19-24 enable Qorvo to create BAW filters of a high level of quality that drives its business. Trade Secrets 19-24 took Qorvo several years, and cost Qorvo millions of dollars to create. By misappropriating Trade Secrets 19-24, Akoustis obtained an unfair advantage, including but not limited to obtaining a head start in producing and offering competing BAW filters accounting for the same quality considerations. Moreover, Akoustis avoided the great costs and expenses of developing its own testing methods and criteria through trial and error.

91. Akoustis misappropriated Trade Secrets 19-24 by improperly acquiring, accessing, disclosing and using such trade secrets for its own benefit. Upon information and belief, Akoustis' misappropriation is continuing. Akoustis and its employees knew or had reason to know that Trade Secrets 19-24 were acquired by improper means, including through former Qorvo employees who improperly disclosed Trade Secrets 19-24 withing Akoustis and to its other employees (including officers and other executives at Akoustis). [REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

92. Upon information and belief, in addition to acquiring, accessing, and disclosing Qorvo's trade secret information, Akoustis knowingly used and continues to use the information to advance its business. [REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

Trade Secrets 25-29 (Exhibits E-1, E-2)

93. Through years of research and development, and at great expense, Qorvo developed trade secrets related to the manufacture and production of semiconductor devices, including but not limited to the inspection of products such as BAW filters. These trade secrets existed in written form at least as early as April 26, 2017, including as reflected in a confidential-marked Qorvo document attached hereto as Exhibit E-1 and bearing the bates label QORVO_00034983-5001.

94. Qorvo trade secrets contained in Exhibit E-1 include, but are not limited to: (a) details regarding [REDACTED] Qorvo's inspection procedures for its semiconductor devices ("Trade Secret 25"), see Exhibit E-1 at QORVO_00034984-985; (b) details regarding Qorvo's inspection procedures ("Trade Secret 26"), see Exhibit E-1 at QORVO_00034985-87; (c) [REDACTED] procedures for Qorvo's inspection procedures ("Trade Secret 27"), see Exhibit E-1 at QORVO_00034985-87; (d) [REDACTED] procedures for Qorvo's inspection procedures ("Trade Secret 28"), see Exhibit E-1 at QORVO_00034986-88;

[REDACTED]

and (e) criteria for accepting or rejecting products according to Qorvo's inspection procedure ("Trade Secret 29"), see Exhibit E-1 at QORVO_00034988-5000.

95. Qorvo has taken and continues to take commercially reasonable steps to protect Trade Secrets 25-29. As set forth in paragraphs 1 through 29, Qorvo protects its trade secrets at least in part by maintaining the confidentiality of documents containing such information.

96. As a result, none of Trade Secrets 25-29 were or are generally known or readily ascertainable, nor can they be obtained through the public domain. Nor could Trade Secrets 25-29 be determined through reverse engineering. Trade Secrets 25-29 derive independent economic value from not being generally known to, and not being readily ascertainable through proper means by, entities who can obtain economic value from the disclosure or use of the information. Trade Secrets 25-29 create competitive value for Qorvo by enabling its manufacturing team to efficiently and accurately inspect its products and determine whether such products have been manufactured to an acceptable standard. This information, in turn, informs Qorvo's engineers in making modifications and improvements to the manufacturing process to avoid the issues leading to unacceptable products. This enables Qorvo to, among other things, create more reliable and more robust filters that can perform with greater resilience when deployed in real-world environments, and gives Qorvo a competitive edge in the industry. In short, Trade Secrets 25-29 enable Qorvo to create products, including BAW filters, of a high level of quality that drives its business and sets it apart from its competitors. Trade Secrets 25-29 took Qorvo several years, and cost Qorvo millions of dollars, to create. By misappropriating Trade Secrets 25-29, Akoustis obtained an advantage, including but not limited to obtaining a head start in producing and offering competing BAW filters and other products accounting for the same quality considerations. Moreover,

[REDACTED]

Akoustis avoided the great costs and expenses of developing its own inspection procedures and criteria through trial and error.

97. Akoustis misappropriated Trade Secrets 25-29 by improperly acquiring, accessing, disclosing and using such trade secrets for its own benefit. Upon information and belief, Akoustis' misappropriation is continuing. Akoustis and its employees knew or had reason to know that Trade Secrets 25-29 were acquired by improper means, including through at least one former Qorvo employee who improperly disclosed Trade Secrets 25-29 to Akoustis personnel. [REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

98. Upon information and belief, in addition to acquiring, accessing, and disclosing Qorvo's trade secret information, Akoustis knowingly used and continues to use the information to advance its business.

Trade Secrets 30-32 (Exhibits F-1, F-2)

99. Through years of research and development, and at great expense, Qorvo developed trade secrets related to implementation requirements for [REDACTED] of assembled

[REDACTED]

products, including BAW filters. These trade secrets existed in written form at least as early as February 13, 2017, including as reflected in a confidential marked Qorvo document attached hereto as Exhibit F-1, [REDACTED]

[REDACTED] Qorvo trade secrets contained in Exhibit F-1 include, but are not limited to: (a) details relating to Qorvo's procedure for [REDACTED], including [REDACTED] [REDACTED] ("Trade Secret 30"), see Exhibit F-1 at QORVO_00035121; (b) details and instructions regarding performing Qorvo's [REDACTED] ("Trade Secret 31"), see Exhibit F-1 at QORVO_00035121-123; and (c) details regarding the [REDACTED] [REDACTED] ("Trade Secret 32"), see Exhibit F-1 at QORVO_00035122-123.

100. Qorvo has taken and continues to take commercially reasonable steps to protect Trade Secrets 30-32. As set forth in paragraphs 1 through 29, Qorvo protects its trade secrets at least in part by maintaining the confidentiality of documents containing such information.

101. As a result, none of Trade Secrets 30-32 were or are generally known or readily ascertainable, nor can they be obtained through the public domain. Nor could Trade Secrets 30-32 be determined through reverse engineering. Trade Secrets 30-32 derive independent economic value from not being generally known to, and not being readily ascertainable through proper means by, entities who can obtain economic value from the disclosure or use of the information. Trade Secrets 30-32 create competitive value for Qorvo by enabling assembly, manufacturing, and engineering teams to produce BAW filters and other products that are more reliable and more robust and can perform with greater resilience when deployed in real-world environments, and giving Qorvo a competitive edge in the industry. In short, Trade Secrets 30-32 enable Qorvo to

[REDACTED]

create BAW filters of a high level of quality that drives its business. Trade Secrets 30-32 took Qorvo several years, and cost Qorvo millions of dollars, to create. By misappropriating Trade Secrets 30-32, Akoustis obtained an unfair advantage, including but not limited to obtaining a head start in producing and offering competing BAW filters accounting for the same quality considerations. Moreover, Akoustis avoided the great costs and expenses of developing its own reflow preconditioning procedure through trial and error.

102. Akoustis misappropriated Trade Secrets 30-32 by improperly acquiring, accessing, disclosing and using such trade secrets for its own benefit. Upon information and belief, Akoustis' misappropriation is continuing. Akoustis and its employees knew or had reason to know that Trade Secrets 30-32 were acquired by improper means, including through former Qorvo employees who improperly disclosed Trade Secrets 30-32 within Akoustis and to its employees. [REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

103. Upon information and belief, Mr. Dry doctored Qorvo's confidential document in attempt to conceal his misappropriation and use of Qorvo confidential information. As stated above, Mr. Dry is a former Qorvo employee who knew or had reason to know that Exhibits F-1 and F-2 contained Qorvo's trade secrets. Moreover, because Exhibit F-1 was marked as Qorvo's "Confidential & Proprietary Information" and was, or at least a version of it was, shared by former Qorvo employees having obligations of confidentiality to Qorvo, Akoustis and its employees knew or should have known that Exhibits F-1 and/or F-2 were derived from or through a person who owed a duty of non-disclosure to Qorvo.

104. Upon information and belief, in addition to acquiring, accessing, and disclosing Qorvo's trade secret information, upon information and belief, Akoustis knowingly used and continues to use the information to advance its business. [REDACTED]

[REDACTED]

Trade Secrets 33-36 (Exhibits G-1, G-2, G-3)

105. Through years of research and development, and at great expense, Qorvo developed trade secrets related to implementation requirements for [REDACTED] of assembled products, including BAW filters. These trade secrets existed in written form at least as early as March 24, 2017, including as reflected in a confidential-marked Qorvo document attached hereto as Exhibit G-1, [REDACTED]

[REDACTED]. Qorvo trade secrets contained in Exhibit G-1 include, but are not limited to: (a) details relating to Qorvo's procedure for [REDACTED], including [REDACTED]

[REDACTED] ("Trade Secret 33"); see Exhibit G-1 at QORVO_00034926-30; (b) details concerning criteria [REDACTED]

[REDACTED]

[REDACTED] (“Trade Secret 34”), *see* Exhibit G-1 at QORVO_00034929; (c) details regarding [REDACTED]

[REDACTED] (“Trade Secret 35”), *see* Exhibit G-1 QORVO_00034927-28; and (d) customer details, including [REDACTED]

[REDACTED] (“Trade Secret 36”), *see* Exhibit G-1 at QORVO_00034929-30.

106. Qorvo has taken and continues to take commercially reasonable steps to protect Trade Secrets 33-36. As set forth in paragraphs 1 through 29, Qorvo protects its trade secrets at least in part by maintaining the confidentiality of documents containing such information.

107. As a result, none of Trade Secrets 33-36 were or are generally known or readily ascertainable, nor can they be obtained through the public domain. Nor could Trade Secrets 33-36 be determined through reverse engineering. Trade Secrets 33-36 derive independent economic value from not being generally known to, and not being readily ascertainable through proper means by, entities who can obtain economic value from the disclosure or use of the information. Trade Secrets 33-36 create competitive value for Qorvo by enabling assembly, manufacturing, and engineering teams to produce BAW filters and other products that are more reliable and more robust and can perform with greater resilience when deployed in real-world environments, and giving Qorvo a competitive edge in the industry. In short, Trade Secrets 33-36 enable Qorvo to create BAW filters of a high level of quality that drives its business. Trade Secrets 33-36 took Qorvo several years, and cost Qorvo millions of dollars, to create. By misappropriating Trade Secrets 33-36, Akoustis obtained an unfair advantage, including but not limited to obtaining a head start in its ability to produce and offer competing BAW filters accounting for the same quality

[REDACTED]

[REDACTED]

considerations. Moreover, Akoustis avoided the great costs and expenses of developing its own reflow preconditioning procedure through trial and error.

108. Akoustis misappropriated Trade Secrets 33-36 by improperly acquiring, accessing, disclosing and using such trade secrets for its own benefit. Upon information and belief, Akoustis' misappropriation is continuing. Akoustis and its employees knew or had reason to know that Trade Secrets 33-36 were acquired by improper means, including through former Qorvo employees who improperly disclosed Trade Secrets 33-36 within Akoustis and to its employees. [REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

109. Upon information and belief, Mr. Dry doctored Qorvo's confidential document in attempt to conceal his misappropriation and use of Qorvo confidential information. As stated above, Mr. Dry is a former Qorvo employee who knew or had reason to know that Exhibits G-1 and G-3 contained Qorvo's trade secrets. Moreover, because Exhibit G-1 was marked as Qorvo's

[REDACTED]

“Confidential & Proprietary Information” and was, or at least a version of it was, shared by former Qorvo employees having obligations of confidentiality to Qorvo, Akoustis and its employees knew or should have known that Exhibits G-1 and/or G-3 were derived from or through a person who owed a duty of non-disclosure to Qorvo. In addition to acquiring, accessing, and disclosing Qorvo’s trade secret information, upon information and belief, Akoustis knowingly used and continues to use the information to advance its business. [REDACTED]

[REDACTED]

Trade Secrets 37-46 (Exhibits H-1, H-2, H-3)

110. Through years of research and development, and at great expense, Qorvo developed trade secrets related to its QPF4551 product, [REDACTED]. [REDACTED]. These trade secrets existed in written form at least as early as October 5, 2017, including as reflected in a confidential-marked Qorvo document attached hereto as Exhibit H-1 and bearing the bates label QORVO_00059418 (“Exhibit H-1”). Qorvo trade secrets contained in Exhibit H-1 include, but are not limited to: (a) product [REDACTED] details (“Trade Secret 37”), see Exhibit H-1 at Overview sheet; (b) product specifications (“Trade Secret 38”), see Exhibit H-1; and (c) detailed product technical information relating to the product, including [REDACTED] (“Trade Secret 39”), see Exhibit H-1.

111. Qorvo has taken and continues to take commercially reasonable steps to protect Trade Secrets 37-39. As set forth in paragraphs 1 through 29, Qorvo protects its trade secrets at least in part by maintaining the confidentiality of documents containing such information.

112. As a result, none of Trade Secrets 37-39 were or are generally known or readily ascertainable, nor can they be obtained through the public domain. Nor could Trade Secrets 37-39 be determined through reverse engineering. Trade Secrets 37-39 derive independent economic

[REDACTED]

value from not being generally known to, and not being readily ascertainable through proper means by, entities who can obtain economic value from the disclosure or use of the information. Trade Secrets 37-39 create competitive value for Qorvo by enabling it design, manufacture, test, price, and sell its product according to acceptable standards. In short, Trade Secrets 37-39 enable Qorvo to create products of a high level of quality that drives its business. Trade Secrets 37-39 took Qorvo several years, and cost Qorvo millions of dollars, to create. By misappropriating Trade Secrets 37-39, Akoustis obtained an advantage, including but not limited to obtaining a head start in producing and offering products accounting for the same quality considerations. Moreover, Akoustis avoided the great costs and expenses of developing its own products.

113. Akoustis misappropriated Trade Secrets 37-39 by improperly acquiring, accessing, disclosing and using such trade secrets for its own benefit. Upon information and belief, Akoustis' misappropriation is continuing. Akoustis and its employees knew or had reason to know that Trade Secrets 37-39 were acquired by improper means, including through at least one former Qorvo employee who improperly disclosed Trade Secrets 37-39 to Akoustis personnel. [REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

114. Mr. Houlden's email attached another confidential-marked Qorvo document attached hereto as Exhibit H-3 and bearing the bates label AKTS_00145342 ("Exhibit H-3"). Qorvo invested great resources to create trade secrets reflected in this Product Review Document template that describes the Qorvo QPF7250 product, as shown in Exhibit H-3, to organize necessary information for the design, test, development, and sales of its products. Such trade secrets include, but are not limited to: (a) a description of how the [REDACTED] Document is to be used, in the [REDACTED] sheet ("Trade Secret 40"); (b) a description of the [REDACTED] product, including its performance and how it compares to competing products, [REDACTED] ("Trade Secret 41"); (c) key features and differentiators of the product, in the [REDACTED] ("Trade Secret 42"); (d) a [REDACTED] target for the product, [REDACTED] ("Trade Secret 43"); (e) details regarding the product's specifications and analysis of [REDACTED] ("Trade Secret 44"); (f) details regarding test specifications for the product, [REDACTED] ("Trade Secret 45"); and (g) an applications schematic, block diagram and pin layout, and logic table for the [REDACTED] ("Trade Secret 46").

115. Qorvo has taken and continues to take commercially reasonable steps to protect Trade Secrets 40-46. As set forth in paragraphs 1 through 29, Qorvo protects its trade secrets at least in part by maintaining the confidentiality of documents containing such information.

116. As a result, none of Trade Secrets 40-46 were or are generally known or readily ascertainable, nor can they be obtained through the public domain. Nor could Trade Secrets 40-46

be determined through reverse engineering. Trade Secrets 40-46 derive independent economic value from not being generally known to, and not being readily ascertainable through proper means by, entities who can obtain economic value from the disclosure or use of the information. Trade Secrets 40-46 create competitive value for Qorvo by enabling it design, manufacture, test, price, and sell its product according to acceptable standards. In short, Trade Secrets 40-46 enable Qorvo to create products of a high level of quality that drives its business. Trade Secrets 40-46 took Qorvo several years, and cost Qorvo millions of dollars, to create. By misappropriating Trade Secrets 40-46, Akoustis obtained an unfair advantage, including but not limited to obtaining a head start in producing and offering products accounting for the same quality considerations. Moreover, Akoustis avoided the great costs and expenses of developing its own products.

117. Akoustis misappropriated Trade Secrets 40-46 by improperly acquiring, accessing, disclosing and using such trade secrets for its own benefit. Upon information and belief, Akoustis' misappropriation is continuing. Akoustis and its employees knew or had reason to know that Trade Secrets 40-46 were acquired by improper means, including through former at least one former Qorvo employee who improperly disclosed Trade Secrets 40-46 with Akoustis personnel. At least because these Ex-Qorvo employees had access to Qorvo documents and trade secrets during their employment with Qorvo, such Ex-Qorvo employees had a specific opportunity to acquire the same for disclosure or use without the express or implied consent or authority of Qorvo. Furthermore, at least because Akoustis recruited and hired these employees (those who had accessed and taken such Qorvo documents), Akoustis, as their new employer, had a specific opportunity to improperly obtain and use the documents, and the trade secrets within them, without the express or implied consent or authority of Qorvo.

[REDACTED]

118. [REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

119. Upon information and belief, in addition to acquiring, accessing, and disclosing Qorvo's trade secret information, Akoustis knowingly used and continues to use the information to advance its business.

Trade Secrets 47-51 (Exhibits I-1, I-2, I-3, I-4, I-5)

120. Through years of research and development, and at great expense, Qorvo developed trade secrets relating to its QPD2593 product. These trade secrets existed in written form at least as early as August 18, 2016, including as reflected in a confidential-marked Qorvo document attached hereto as Exhibit I-1 and bearing the bates label QORVO_00035126-132. Qorvo trade secrets contained in Exhibit I-1 include, but are not limited to: (a) assembly instructions for the QPD2593 product ("Trade Secret 47"), *see* Exhibit I-1 at QORVO_00035126-132; (b) a die specification and die information for the product ("Trade Secret 48"), *see* Exhibit I-1 at QORVO_00035127; (c) instructions for [REDACTED] for the product ("Trade Secret 49"), *see* Exhibit I-1 at QORVO_00035128; (d) a diagram showing the location of components of the product ("Trade Secret 50"), *see* Exhibit I-1 at QORVO_00035129-30; and (d) information

relating to the wirebonding specification for the product (“Trade Secret 51”), *see* Exhibit I-1 at QORVO_00035131.

121. Qorvo has taken and continues to take commercially reasonable steps to protect Trade Secrets 47-53. As set forth in paragraphs 1 through 29, Qorvo protects its trade secrets at least in part by maintaining the confidentiality of documents containing such information.

122. As a result, none of Trade Secrets 47-51 were or are generally known or readily ascertainable, nor can they be obtained through the public domain. Nor could Trade Secrets 47-51 be determined through reverse engineering. Trade Secrets 47-51 derive independent economic value from not being generally known to, and not being readily ascertainable through proper means by, entities who can obtain economic value from the disclosure or use of the information. Trade Secrets 47-51 create competitive value for Qorvo by enabling it to assemble its product according to acceptable standards. In short, Trade Secrets 47-51 enable Qorvo to create a product of a high level of quality that drives its business. Trade Secrets 47-51 took Qorvo several years, and cost Qorvo millions of dollars, to create. By misappropriating Trade Secrets 47-51, Akoustis obtained an unfair advantage, including but not limited to obtaining a head start in producing and offering products accounting for the same quality considerations. Moreover, Akoustis avoided the great costs and expenses of developing its own products.

123. Akoustis misappropriated Trade Secrets 47-51 by improperly acquiring, accessing, disclosing and using such trade secrets for its own benefit. Upon information and belief, Akoustis’ misappropriation is continuing. Akoustis and its employees knew or had reason to know that Trade Secrets 47-51 were acquired by improper means, including through at least one former Qorvo employee who improperly disclosed Trade Secrets 47-51 with Akoustis personnel. At least because these Ex-Qorvo employees had access to Qorvo documents and trade secrets during their

[REDACTED]

employment with Qorvo, such Ex-Qorvo employees had a specific opportunity to acquire the same for disclosure or use without the express or implied consent or authority of Qorvo. Furthermore, at least because Akoustis recruited and hired these employees (those who had accessed and taken such Qorvo documents), Akoustis, as their new employer, had a specific opportunity to improperly obtain and use the documents, and the trade secrets within them, without the express or implied consent or authority of Qorvo.

124. [REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

Trade Secrets 52-63 (Exhibits J-1, J-2, J-3)

125. Through years of research and development, and at great expense, Qorvo developed trade secrets relating to detailed domestic and international compensation and business structure intelligence which [REDACTED]. This intelligence included detailed promotion criteria for each job title through years of training employees and human resources analyses. These trade secrets existed in written form at least as early as May 29, 2017, including as reflected in a confidential-marked Qorvo document attached hereto as Exhibit J-1 and bearing the bates label QORVO_00039655. Qorvo trade secrets contained in Exhibit J-1 include, but are not limited to: (a) [REDACTED] of technical skillsets necessary for each job title (“Trade Secret 52”); see Exhibit J-1, “Job descriptions” sheet; (b) [REDACTED] necessary to determine promotion eligibility (“Trade Secret 53”), see Exhibit J-1, [REDACTED]; (c) Qorvo’s target salary ranges for each job title in the United States (“Trade Secret 54”), see Exhibit J-1, [REDACTED]; (d) Qorvo’s target salary ranges for each job title [REDACTED] (“Trade Secret 55”), see Exhibit J-1, [REDACTED]; (e) Qorvo’s target salary ranges for each job title [REDACTED] (“Trade Secret 56”), see Exhibit J-1 at “[REDACTED]” and “[REDACTED]”; (f) Qorvo’s target salary ranges for each job title [REDACTED] (“Trade Secret 57”), see Exhibit J-1 at [REDACTED] sheets; (g) Qorvo’s target salary ranges for each job title [REDACTED] (“Trade Secret 58”), see Exhibit J-1 at [REDACTED]; (h) Qorvo’s target salary ranges for each job title [REDACTED] (“Trade Secret 59”), see Exhibit J-1 at [REDACTED]; (i) Qorvo’s target salary ranges for each job title [REDACTED] (“Trade Secret 60”), see Exhibit J-1 at [REDACTED]; (j) Qorvo’s target salary ranges for each job title [REDACTED] (“Trade Secret 61”), see Exhibit J-1 at [REDACTED]

[REDACTED]

[REDACTED] (k) Qorvo's target salary ranges for each job title [REDACTED] ("Trade Secret 62"), see Exhibit J-1 at [REDACTED]; and (l) Qorvo's target salary ranges for persons hired through [REDACTED] ("Trade Secret 63"), see Exhibit J-1 [REDACTED]

126. Qorvo has taken and continues to take commercially reasonable steps to protect Trade Secrets 52-63. As set forth in paragraphs 1 through 29, Qorvo protects its trade secrets at least in part by maintaining the confidentiality of documents containing such information.

127. As a result none of Trade Secrets 52-63 were or are generally known or readily ascertainable, nor can they be obtained through the public domain. Nor could Trade Secrets 52-63 be determined through reverse engineering. Trade Secrets 52-63 derive independent economic value from not being generally known to, and not being readily ascertainable through proper means by, entities who can obtain economic value from the disclosure or use of the information. Trade Secrets 52-63 create competitive value for Qorvo by enabling Qorvo to strike a delicate balance between [REDACTED]

[REDACTED]

[REDACTED] In short, Trade Secrets 52-63 enable Qorvo to hire and retain highly competent employees who are critical to its business. Trade Secrets 52-63 took Qorvo several years, and cost Qorvo millions of dollars, to create. By misappropriating Trade Secrets 52-63, Akoustis obtained an unfair advantage, including but not limited to obtaining a head start in its ability to create an effective hiring process, internal performance evaluation manuals, and most importantly, effectively poach key Qorvo employees. Moreover, Akoustis avoided the great costs and expenses of losing employees, hiring employees without the necessary skillsets, and promoting ill-prepared employees, all avoidable only through many years of trial and error.

[REDACTED]

128. Akoustis misappropriated Trade Secrets 52-63 by improperly acquiring, accessing, disclosing and using such trade secrets for its own benefit. Upon information and belief, Akoustis' misappropriation is continuing. Akoustis and its employees knew or had reason to know that 52-63 were acquired by improper means, including through former Qorvo employees who improperly disclosed Trade Secrets 52-63 within Akoustis and to its other employees. [REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

Trade Secrets 64-67 (Exhibits K-1, K-2, K-3)

129. Through years of research and development, and at great expense, Qorvo developed trade secrets relating to its branding requirements and guidelines for marking its products [REDACTED]

[REDACTED]. These trade secrets existed in written form at least as early as January 7, 2015, including as reflected in a confidential-marked Qorvo document attached hereto as Exhibit K-1 and bearing the bates label QORVO_00035135-

[REDACTED]

148 (“Exhibit K-1”). Qorvo trade secrets contained in Exhibit K-1 include, but are not limited to guidelines and instructions for laser and ink marking Qorvo’s products, including with respect to

(a) [REDACTED]
[REDACTED] for such markings (“Trade Secret 64”), *see* Exhibit K-1 at QORVO_00035135-148; (b) marking [REDACTED] required by customers and/or to be used in connection with assembly (“Trade Secret 65”), *see* Exhibit K-1 at QORVO_00035135-148; (c) [REDACTED] to create a legible “Qorvo” label (“Trade Secret 66”), *see* Exhibit K-1 at QORVO_00035135-148; and (d) [REDACTED] row configuration and prioritization hierarchy [REDACTED] [REDACTED] (“Trade Secret 67”), *see* Exhibit K-1 at QORVO_00035135-148.

130. Qorvo has taken and continues to take commercially reasonable steps to protect Trade Secrets 64-67. As set forth in paragraphs 1 through 29, Qorvo protects its trade secrets at least in part by maintaining the confidentiality of documents containing such information.

131. As a result, none of Trade Secrets 64-67 were or are generally known or readily ascertainable, nor can they be obtained through the public domain. Nor could Trade Secrets 64-67 be determined through reverse engineering. Trade Secrets 64-67 derive independent economic value from not being generally known to, and not being readily ascertainable through proper means by, entities who can obtain economic value from the disclosure or use of the information. Trade Secrets 64-67 create competitive value for Qorvo [REDACTED]
[REDACTED]
[REDACTED]

[REDACTED]. In short, Trade Secrets 64-67 enable Qorvo to create a product of a high level of quality and usability in the assembly process, that drives Qorvo’s business. Trade Secrets 64-67

[REDACTED]

took Qorvo several years and cost Qorvo tens of thousands of dollars to create. By misappropriating Trade Secrets 64-67, Akoustis obtained an advantage, including but not limited to obtaining a head start in producing and offering products accounting for the same quality and usability considerations. [REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

132. Akoustis misappropriated Trade Secrets 64-67 by improperly acquiring, accessing, disclosing and using such trade secrets for its own benefit. Upon information and belief, Akoustis' misappropriation is continuing. Akoustis and its employees knew or had reason to know that Trade Secrets 64-67 were acquired by improper means, including through at least one former Qorvo employee who improperly disclosed Trade Secrets 64-67 with Akoustis personnel. At least because these Ex-Qorvo employees had access to Qorvo documents and trade secrets during their employment with Qorvo, such Ex-Qorvo employees had a specific opportunity to acquire the same for disclosure or use without the express or implied consent or authority of Qorvo. Furthermore, at least because Akoustis recruited and hired these employees (those who had accessed and taken such Qorvo documents), Akoustis, as their new employer, had a specific opportunity to improperly obtain and use the documents, and the trade secrets within them, without the express or implied consent or authority of Qorvo.

133. [REDACTED]

[REDACTED]

[REDACTED]

[REDACTED] [REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED] Upon information and belief, in addition to acquiring, accessing, and disclosing Qorvo's trade secret information, Akoustis knowingly used and continues to use the information to advance its business.

Trade Secrets 68-85 (Exhibits L-1, L-2, L-3, L-4)

134. Through years of research and development, and at great expense, Qorvo developed trade secrets relating to BAW filters in development, including high-performance, high power Bulk Acoustic Wave (BAW) band-pass filters with extremely steep skirts. [REDACTED]

[REDACTED]

[REDACTED]. Qorvo invested significant resources into developing

[REDACTED]

this BAW filter to enables, among other [REDACTED]

[REDACTED]. The products under development include filters with highly unique power handling capabilities that allow for implementation into high-power applications. The trade secrets concerning these filters, their applications, technical architecture and layout plans, and product roadmaps existed in written form at least as early as January 2019, including as reflected in a confidential-marked Qorvo document attached hereto as Exhibit L-1 and bearing the bates label QORVO_00034746-821. See also the confidential-marked Qorvo document attached hereto as Exhibit L-2 and bearing the bates label QORVO_00034822-896, and containing some overlapping content. Qorvo trade secrets contained in Exhibit L-1 include, but are not limited to Qorvo's: (a) product roadmap for [REDACTED] [REDACTED] ("Trade Secret 68"), see Exhibit L-1 at QORVO_00034758; (b) technical details and architecture of [REDACTED] ("Trade Secret 69"), see Exhibit L-1 at QORVO_00034759; (c) block diagrams and technical details of the [REDACTED] ("Trade Secret 70"), see Exhibit L-1 at QORVO_00034760; (d) block diagrams and technical details of the [REDACTED] ("Trade Secret 71"), see Exhibit L-1 at QORVO_00034761; (e) block diagrams and technical details of the [REDACTED] ("Trade Secret 72"), see Exhibit L-1 at QORVO_00034762; (f) study results, targets, and technical data concerning [REDACTED] ("Trade Secret 73"), see Exhibit L-1 at QORVO_00034764; (g) [REDACTED] performance and test data/results in connection with [REDACTED] [REDACTED] [REDACTED] [REDACTED] ("Trade Secret 74"), see Exhibit L-1 at QORVO_00034766; (h) market intelligence concerning [REDACTED] [REDACTED] ("Trade Secret 75"), see Exhibit L-1 at QORVO_00034770-73; (i) architectural details, block diagrams for [REDACTED]

[REDACTED]

[REDACTED] (“Trade Secret 76”), *see* Exhibit L-1 at QORVO_00034775-76; (j) architectural details for [REDACTED] (“Trade Secret 77”), *see* Exhibit L-1 at QORVO_00034777, 790; (k) performance data, test data, and block diagrams for [REDACTED] (“Trade Secret 78”), *see* Exhibit L-1 at QORVO_00034780-83; (l) product roadmap for [REDACTED] (“Trade Secret 79”), *see* Exhibit L-1 at QORVO_00034784; (m) product roadmap for [REDACTED] (“Trade Secret 80”), *see* Exhibit L-1 at QORVO_00034789; (n) technical details and architecture of the [REDACTED] (“Trade Secret 81”), *see* Exhibit L-1 at QORVO_00034790-91; (o) data and technical data concerning [REDACTED] (“Trade Secret 82”), *see* Exhibit L-1 at QORVO_00034792-97; (p) product roadmap for [REDACTED] (“Trade Secret 83”), *see* Exhibit L-1 at QORVO_00034799; (r) block diagrams and product features for products under development, and identification of [REDACTED] (“Trade Secret 84”), *see* Exhibit L-1 at QORVO_00034800-808; (r) product data and related block diagrams (“Trade Secret 85”), *see* Exhibit L-1 at QORVO_00034816-820.

135. Qorvo has taken and continues to take commercially reasonable steps to protect Trade Secrets 68-85. As set forth in paragraphs 1 through 29, Qorvo protects its trade secrets at least in part by maintaining the confidentiality of documents containing such information.

136. As a result, none of Trade Secrets 68-85 were or are generally known or readily ascertainable, nor can they be obtained through the public domain. Nor could Trade Secrets 68-85 be determined through reverse engineering. Trade Secrets 68-85 derive independent economic value from not being generally known to, and not being readily ascertainable through proper means

[REDACTED]

[REDACTED]

by, entities who can obtain economic value from the disclosure or use of the information. Trade Secrets 68-85 create competitive value for Qorvo by enabling it to mark and brand its [REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]. In short, Trade Secrets 68-85 enable [REDACTED]

[REDACTED]. Trade Secrets 68-85 took Qorvo several years and cost Qorvo millions of dollars to create. By misappropriating one or more of Trade Secrets 68-85, Akoustis obtained an advantage, including but not limited to obtaining a head start in producing and offering products accounting for the same quality, performance and usability considerations.

137. Akoustis misappropriated Trade Secrets 68-85 by improperly acquiring, accessing, disclosing and using such trade secrets for its own benefit. Upon information and belief, Akoustis' misappropriation is continuing. Akoustis and its employees knew or had reason to know that Trade Secrets 68-85 were acquired by improper means, including through at least one former Qorvo employee who improperly disclosed one or more of Trade Secrets 68-85 with Akoustis personnel. At least because these Ex-Qorvo employees had access to Qorvo documents and trade secrets during their employment with Qorvo, such Ex-Qorvo employees had a specific opportunity to acquire the same for disclosure or use without the express or implied consent or authority of Qorvo. Furthermore, at least because Akoustis recruited and hired these employees (those who had accessed and taken such Qorvo documents), Akoustis, as their new employer, had a specific opportunity to improperly obtain and use the documents, and the trade secrets within them, without the express or implied consent or authority of Qorvo.

[REDACTED]

138. [REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

Trade Secret 86 (Exhibits M-1, M-2)

139. Through years of research and development, and at great expense, Qorvo developed trade secrets relating to BAW filter simulation and performance data that informed its research, development, and competitive analysis in the BAW filter space. Qorvo invested significant resources into generating such data, identifying parameters/variables to modify to best understand underlying products associated with this data, and interpreting this data to inform the way it refines its products. This data is highly sensitive. The trade secrets reflected in this data existed in the

[REDACTED]

form of, e.g., [REDACTED] generated by simulation software, at least as early as January 7, 2019. [REDACTED]

[REDACTED] The trade secrets arising out of Qorvo generated [REDACTED] are collectively referred to herein as “Trade Secret 86.” [REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

140. Qorvo has taken and continues to take commercially reasonable steps to protect Trade Secret 86. As set forth in paragraphs 1 through 29, Qorvo protects its trade secrets at least in part by maintaining the confidentiality of documents containing such information.

141. As a result, Trade Secret 86 was not or is not generally known or readily ascertainable, nor can it be obtained through the public domain. Trade Secret 86 derives independent economic value from not being generally known to, and not being readily ascertainable through proper means by, entities who can obtain economic value from the disclosure or use of the information. Trade Secret 86 creates competitive value for Qorvo by enabling it to evaluate its products at various points throughout the development process, and under different conditions of unique interest, and refine their BAW filter designs to produce the most reliable and highest performance filters relative to their competition. In short, Trade Secret 86 enables Qorvo to create a product of a high level of quality and usability that drives Qorvo’s business. Trade Secret 86 took Qorvo considerable time and financial resource to create. By misappropriating

[REDACTED]

Trade Secret 86, Akoustis obtained an advantage, including but not limited to obtaining a head start in their competitive analysis and production of BAW filter products, among others, that account for the same performance and usability considerations that Qorvo learned through trial and error over an extensive period of time.

142. Akoustis misappropriated Trade Secret 86 by improperly acquiring, accessing, disclosing and using such trade secrets for its own benefit. Upon information and belief, Akoustis' misappropriation is continuing. Akoustis and its employees knew or had reason to know that Trade Secret 86 were acquired by improper means, including through at least one former Qorvo employee who improperly disclosed Trade Secret 86 with Akoustis personnel. At least because these Ex-Qorvo employees had access to Qorvo documents and trade secrets during their employment with Qorvo, such Ex-Qorvo employees had a specific opportunity to acquire the same for disclosure or use without the express or implied consent or authority of Qorvo. Furthermore, at least because Akoustis recruited and hired these employees (those who had accessed and taken such Qorvo documents), Akoustis, as their new employer, had a specific opportunity to improperly obtain and use the documents, and the trade secrets within them, without the express or implied consent or authority of Qorvo.

143. [REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

Additional Trade Secrets (Exhibits N-1)

144. In addition to enumerated Trade Secrets 1–86, Qorvo has developed many other highly valuable trade secrets (“Additional Trade Secrets”). Upon information and belief, Akoustis has improperly acquired, accessed, disclosed and/or used such Additional Trade Secrets in advancing its own business interests. Examples of this are attached hereto as Exhibit N-1, which includes documents bearing the bates labels: AKTS_00156270, AKTS_00117498, AKTS_00138229, AKTS_00138271, AKTS_00141631, AKTS_00151319, AKTS_00143669, AKTS_00145340, AKTS_00146974, AKTS_00146974, AKTS_00167236; AKTS_00198676; AKTS_00198680; AKTS_00198699; AKTS_00198718; AKTS_00198736; AKTS_00198755; AKTS_00198821; AKTS_00198836; AKTS_00198854; AKTS_00198873; AKTS_00198944; AKTS_00199018; AKTS_00199032; AKTS_00199035; AKTS_00199038; AKTS_00199040; AKTS_00199042; AKTS_00199107; AKTS_00199110; AKTS_00199113 (collectively, “Exhibit N-1”).

145. Qorvo has taken and continues to take commercially reasonable steps to protect the Additional Trade Secrets, including as set forth in paragraphs 1 through 29.

146. As a result the Additional Trade Secrets are not generally known or readily ascertainable, nor can they be obtained through the public domain. Nor could the Additional Trade Secrets be determined through reverse engineering. The Additional Trade Secrets derive independent economic value from not being generally known to, and not being readily

[REDACTED]

[REDACTED]

ascertainable through proper means by, entities who can obtain economic value from the disclosure or use of the information. The Additional Trade Secrets create competitive value for Qorvo by, *inter alia*, enabling Qorvo to meet or exceed performance objective and requirements, to accelerate the research and development cycle, to avoid errors in future product builds, to reduce the amount of time between design and manufacture, to reduce waste and other expenditure of resources to achieve profitability targets, to attract and retain talent, to maintain the product quality that customers have come to associate with the “Qorvo” brand, and to isolate the reason why certain defects arise or certain failures are experienced with BAW filters. If misappropriated, a competitor could (and Akoustis did) avoid the great costs and expenses resulting from running a BAW filter business without these Additional Trade Secrets.

147. Upon information and belief, Akoustis misappropriated the Additional Trade Secrets by improperly acquiring, accessing, disclosing and/or using such trade secrets for its own benefit. Upon information and belief, Akoustis’ misappropriation is continuing. Akoustis and its employees knew or had reason to know that the Additional Trade Secrets were Qorvo’s confidential information, at least on account of the “Confidential” and/or “Proprietary” marking on the documents containing these Additional Trade Secrets, and/or the nature of the content included within these documents. Moreover, Akoustis and its employees knew or had reason to know that the Additional Trade Secrets were acquired by improper means, including through former Qorvo employees who improperly shared such Additional Trade Secrets within Akoustis, knowing that the information was not authorized for such use.

148. Accordingly, Akoustis has improperly built its 3-7 GHz BAW RF filter portfolio, and created related business opportunities, based on the Qorvo BAW Proprietary Information and confidential information, including trade secrets.

149. The Akoustis employees who misappropriated Qorvo's trade secrets knew or had reason to know that the trade secrets, including confidential-marked documents, were acquired by improper means. For example, the Akoustis employees knew that they were not authorized to take confidential Qorvo documents and keep those documents after their employment with Qorvo terminated and knew that doing so violated and/or breached one or more agreements with Qorvo, among other things. In addition, Akoustis employees, including but not limited to Rohan Houlden, in soliciting Qorvo employees to divulge Qorvo confidential documents and information, knew or should have known that they were inducing Qorvo employees to breach their confidentiality agreements with Qorvo. Further, the Akoustis employees acquired confidential-marked Qorvo documents by theft, including by emailing the documents to a personal email account or downloading the documents from Qorvo's computer systems without authorization to use them in the intended manner or the manner in which they were used. Moreover, the Akoustis employees disclosed and/or used Qorvo's trade secrets without Qorvo's consent and in violation and/or breach of one or more agreements with Qorvo.

150. Akoustis and Qorvo are direct competitors in the BAW filter market. Qorvo's trade secrets, including confidential-marked documents, were directly transferable for use by Akoustis. By way of non-limiting example, Akoustis took certain confidential-marked Qorvo documents as described above and re-purposed them to be used by Akoustis for essentially the same purpose for which they were used within Qorvo, including for the development and related activities of competing BAW filters.

151. Akoustis' wrongful conduct alleged herein by their misappropriation of Qorvo's trade secrets has and will continue unless enjoined and restrained by this Court, and will cause great and irreparable injury to Qorvo's business, and it could cause Akoustis to have improper

advantages, positions, and rights in the marketplace to Qorvo's detriment. Absent injunctive relief, Akoustis' further disclosure and use of Qorvo's trade secrets could irreparably harm Qorvo.

152. By using the Qorvo BAW Proprietary Information and other confidential information obtained according to the preceding paragraphs, Akoustis has gained an unfair advantage, enabling Akoustis to misappropriate trade secrets and to enter the BAW filter market more quickly than it could have without the Qorvo BAW Proprietary Information and/or other proprietary or confidential information of Qorvo.

153. On information and belief, Akoustis has used, continues to use, has inevitably used, or will inevitably use, the Qorvo BAW Proprietary Information obtained through Ex-Qorvo BAW Employees and confidential Qorvo documents (including the trade secrets reflected therein) in connection with Akoustis' design, development, manufacture, and qualification of its BAW filters.

E. Akoustis Engages in False Advertising and False Patent Marking By Falsely Describing its BAW Filters as "Single-Crystal"

154. Akoustis' unfair competition has also extended to making false and misleading statements about the characteristics of its BAW filters in an attempt to derive an unfair competitive advantage over competing products including Qorvo products.

155. Akoustis repeatedly promotes and asserts in advertisements, marketing and related materials including investor brochures that its BAW filters use "single-crystal" piezoelectric layers, which it refers to as "XBAW™ technology."

156. For example, this alleged differentiation is prominent on Akoustis' website blog, as shown below (see AKOUSTIS GOES LIVE – AKOUSTIS TECHNOLOGIES, <https://akoustis.com/akoustis-goes-live/> (last visited Sept. 14, 2021)).

Pioneering Next-Generation Materials

At the core of our success lies the conceptualization, development, and manufacturing capabilities of patented XBAW™ technology and our high-purity piezoelectric materials.

Our next-gen materials allow for single-crystal bulk acoustic wave (BAW) high-band RF filters which utilize our advanced resonator-filter process technology to drive higher electro- mechanical coupling wider bandwidth RF filter solutions.

157. Akoustis makes similar claims in its press releases, such as the Press Release related to the AKF-1938 Filter (see AKOUSTIS EXPANDS COMMERCIAL PRODUCT PORTFOLIO WITH NEW 3.8 GHz BAW RF FILTER :: AKOUSTIS TECHNOLOGIES, INC. (AKTS), <https://ir.akoustis.com/news-events/press-releases/detail/90/akoustis-expands-commercial-product-portfolio-with-new-3-8> (last visited Sept. 14, 2021)), as shown below.

The AKF-1938 is a high performance, ultra-small passband 3.8 GHz BAW RF filter designed for use in radar and RF transceiver applications. The filter wafers will be manufactured using Akoustis' [new proprietary XB1 single-crystal BAW manufacturing process](#) which delivers high-performance RF filter solutions for frequencies up to 7 GHz. The AKF-1938 provides

158. Further, in product datasheets for each of the twelve BAW filters identified above that are currently offered for sale on its website (see ALL PRODUCTS – AKOUSTIS TECHNOLOGIES, <https://akoustis.com/products/filters/applications/all-products/> (last visited Sept. 14, 2021)), Akoustis states that the filters utilize “Akoustis’ patented, XBAW technology which provides leading RF filter performance” and that U.S. Patent No. 10,256,786 (the “’786 Patent”) covers that BAW filter.

159. U.S. Patent No. 10,256,786 is titled “Communication Filter Using Single Crystal Acoustic Resonator Devices.” The independent claims of U.S. Patent No. 10,256,786 define

piezoelectric layers, and the disclosure of U.S. Patent No. 10,256,786 and Akoustis' statements make clear that these recited piezoelectric layers are single-crystal.

160. Akoustis relies on its descriptions of its filters as "single-crystal" and the related marking of those filters with U.S. Patent No. 10,256,786 to attract customers and prospective customers in conjunction with its claims that single-crystal piezoelectric materials drive superior performance as compared to polycrystalline technology, such as is used in competitor's BAW filters, including by Qorvo. However, Akoustis' BAW filter products are not "single crystal." They are polycrystalline.

F. Akoustis Engages In Unfair Competition By Promoting Its Infringing, Misappropriated, and Falsely Advertised Products to Qorvo Customers

161. On information and belief, Akoustis has competed with Qorvo and has attempted to persuade Qorvo's customers to replace Qorvo BAW filters with Akoustis' BAW filters that: (i) infringe the Patents-in-Suit; (ii) were developed and manufactured with the Qorvo BAW Proprietary Information and Qorvo's trade secrets; and (iii) are falsely advertised as "single-crystal" and marked with U.S. Patent No. 10,256,786.

V. CLAIMS FOR RELIEF

A. Count I - Infringement of U.S. Patent No. 7,522,018 (the '018 Patent)

162. Qorvo incorporates by reference the allegations of paragraphs 1 through 161 set forth above.

163. Akoustis infringes the '018 Patent under 35 U.S.C. § 271(a), by making, using, selling, or offering for sale the 3-7 GHz BAW RF filter portfolio including, but not limited to its 5.2 GHz RF BAW filters (*i.e.*, AKF-1252), 5.6 GHz RF BAW filters (*i.e.*, AKF-1256), 3.6 GHz CBRS bandpass BAW filters (*i.e.*, AKF-1336), and 3.5 GHz 5G coexistence BAW filters (*i.e.*, AKF-10235) and related products (collectively the "Accused Products"), each of which utilizes

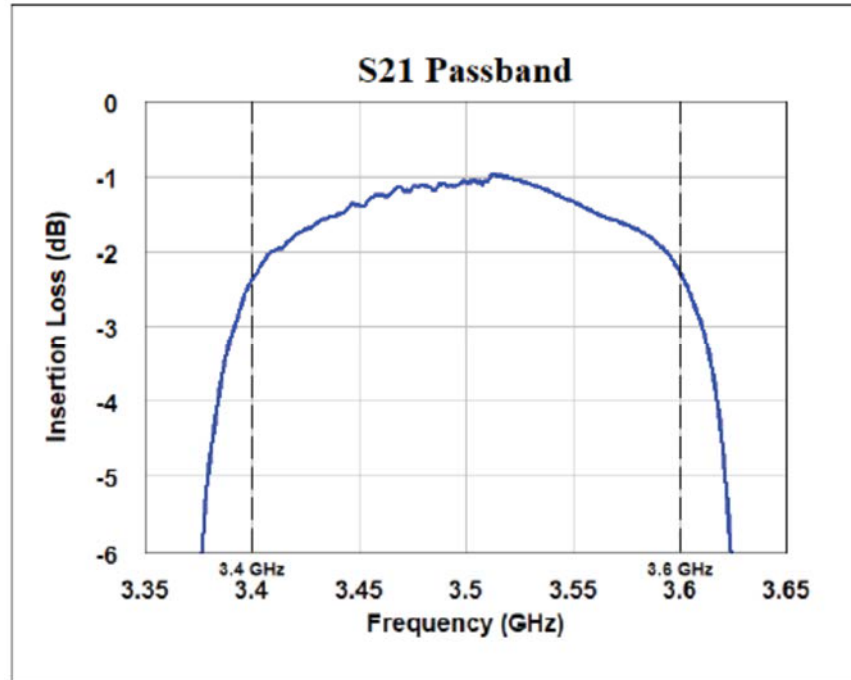
[REDACTED]

the inventions disclosed and claimed in the '018 Patent. On information and belief, each of the Accused Products is of similar design and construction as it pertains to the '018 Patent.

164. Qorvo recently acquired a sample Accused Product, conducted testing and analysis on it, and determined that it meets all of the limitations of one or more claims of the '018 Patent. As an example, Akoustis directly infringes at least independent claim 1 of the '018 Patent. The Accused Products are film bulk acoustic resonators ("FBAR") that have a membrane structure FBAR with a layer structure including a piezoelectric layer and a top and a bottom electrode layer.

165. Like claim 1, the Accused Products are configured with thicknesses of the two electrode layers being unequal, characterized in that the top electrode layer is thinner than the bottom electrode layer. By way of example, the Akoustis AKF-10235 BAW Filter has a top electrode layer that is averages about 110 nm thick and a bottom electrode layer that averages about 129 nm thick. On information and belief, the other Accused Products similarly have piezoelectric layers and top electrode layers that are thinner than their bottom electrode layers.

166. To the extent required by claim 1, the Accused Products have increased filter bandwidth performance. By way of example, increased filter bandwidth performance of the Akoustis AKF-10235 BAW Filter is shown in the below graph taken from the AKF-10235 Data Sheet. On information and belief, the other Accused Products similarly have increased filter bandwidth performance.



167. Thus, Akoustis has infringed, and continues to infringe, at least claim 1 of the '018 Patent in violation of 35 U.S.C. § 271(a), literally or under the doctrine of equivalents, by making, using, selling, and/or offering for sale in the United States, and/or importing into the United States, the Accused Products that are covered by one or more claims of the '018 Patent.

168. On information and belief, Akoustis has been and is inducing infringement of the '018 Patent in violation of 35 U.S.C. § 271(b) at least by actively and knowingly inducing its customers to, literally or under the doctrine of equivalents, make, use, sell, and/or offer for sale in the United States, and/or import into the United States, products incorporating the Accused Products which utilize the inventions disclosed and claimed in the '018 Patent. Specifically, Akoustis induces infringement of the '018 Patent by publishing instructions on its website instructing third parties on how to use the Accused Products. These instructions include recommendations for customers of the Accused Products on how to solder and package the Accused Products as part of customer systems. Akoustis' intent is further demonstrated by PCB

diagrams published on its website that instruct customers on PCB metal, solder mask, and stencil pattern layouts for implementing the Accused Products in infringing systems.

169. As of the introduction of the Accused Products, Akoustis has had knowledge of the '018 Patent and Akoustis' infringement thereof, because the Ex-Qorvo BAW Employees, including former executives of Qorvo who are now executives of Akoustis, knew of the '018 patent and brought that knowledge to Akoustis. Akoustis also has known and knows that its customers' use, sale, and offering for sale of the Accused Products constitutes infringement of at least claim 1 of the '018 Patent for the same reasons set forth above.

170. Akoustis' infringement of the '018 patent is willful, deliberate, and intentional, and Akoustis is acting in reckless disregard of Qorvo's patent rights.

171. Because of Akoustis' infringement of the '018 patent, Qorvo has suffered and will continue to suffer irreparable harm and injury, including monetary damages in an amount to be determined at trial.

172. Unless enjoined, Akoustis, and/or others acting on behalf of Akoustis, will continue their infringing acts, thereby causing additional irreparable injury to Qorvo for which there is no adequate remedy at law.

173. Akoustis' actions render this an exceptional case and entitle Qorvo to attorneys' fees and costs under 35 U.S.C. § 285.

B. Count II - Infringement of U.S. Patent No. 9,735,755 (the '755 Patent)

174. Qorvo incorporates by reference the allegations of paragraphs 1 through 173 set forth above.

175. Akoustis infringes the '755 Patent under 35 U.S.C. § 271(a) by making, using, selling, or offering for sale the 3-7 GHz BAW RF filter portfolio including, but not limited to, its 5.2 GHz RF BAW filters (i.e., AKF-1252), 5.6 GHz RF BAW filters (i.e., AKF-1256), 3.6 GHz

176. Qorvo recently acquired a sample Accused Product, conducted testing and analysis on it, and determined that it meets all of the limitations of one or more claims of the '755 Patent. As an example, Akoustis directly infringes at least independent claim 9 of the '755 Patent. The Accused Products are BAW resonators comprising a piezoelectric layer, a first electrode on a first surface of the piezoelectric layer, a second electrode on a second surface of the piezoelectric layer opposite the first electrode.

178. The Accused Products further include one or more material layers on the second surface of the piezoelectric layer adjacent to the second electrode in an outer region of the BAW resonators. The outer region of the BAW resonators is a region outside of the active region of the BAW resonators. The one or more material layers have a thickness that is n times the thickness (T_{PA}) of the passivation layer within the active region, wherein: n is a value other than 1. For example, the Akoustis AKF-10235 BAW Filter has one or more material layers outside an active region with a thickness of about 1.5 to 1.6 times a thickness of the passivation layer within the active region.

179. In the Accused Products, “n” is such that the outer region of the BAW resonators and the active region of the BAW resonators are acoustically matched in such a manner that the energy of the one or more acoustic wavelengths leaked into the outer region is not excited in the active region.

180. Thus, Akoustis has infringed, and continues to infringe, at least claim 9 of the ’755 Patent in violation of 35 U.S.C. § 271(a), literally or under the doctrine of equivalents, by making, using, selling, and/or offering for sale in the United States, and/or importing into the United States, the Accused Products that are covered by one or more claims of the ’755 Patent.

181. On information and belief, Akoustis has been and is inducing infringement of the ’755 Patent in violation of 35 U.S.C. § 271(b) at least by actively and knowingly inducing its customers to, literally or under the doctrine of equivalents, make, use, sell, and/or offer for sale in the United States, and/or import into the United States, products incorporating the Accused Products which utilize the inventions disclosed and claimed in the ’755 Patent. Specifically, Akoustis induces infringement of the ’755 Patent by publishing instructions on its website instructing third parties on how to use the Accused Products. These instructions include recommendations for customers of the Accused Products on how to solder and package the Accused Products as part of customer systems. Akoustis’ intent is further demonstrated by PCB diagrams published on its website that instruct customers on PCB metal, solder mask, and stencil pattern layouts for implementing the Accused Products in infringing systems.

182. As of the introduction of the Accused Products, Akoustis has had knowledge of the ’755 Patent and Akoustis’ infringement thereof, because the Ex-Qorvo BAW Employees, including former executives of Qorvo who are now executives of Akoustis, knew of the ’755 patent and brought that knowledge to Akoustis. Akoustis also has known and knows that its customers’

use, sale, and offering for sale of the Accused Products constitutes infringement of at least claim 9 of the '755 Patent for the same reasons set forth above.

183. Akoustis' infringement of the '755 patent is willful, deliberate, and intentional, and Akoustis is acting in reckless disregard of Qorvo's patent rights.

184. Because of Akoustis' infringement of the '755 patent, Qorvo has suffered and will continue to suffer irreparable harm and injury, including monetary damages in an amount to be determined at trial.

185. Unless enjoined, Akoustis, and/or others acting on behalf of Akoustis, will continue their infringing acts, thereby causing additional irreparable injury to Qorvo for which there is no adequate remedy at law.

186. Akoustis' actions render this an exceptional case and entitle Qorvo to attorneys' fees and costs under 35 U.S.C. § 285.

C. Count III - Lanham Act False Advertising Under 15 U.S.C. § 1125(a)

187. Qorvo incorporates by reference the allegations of paragraphs 1 through 186 set forth above.

188. Akoustis describes its BAW filters as being "single crystal" construction, which misrepresents the nature, characteristics, and/or qualities of Akoustis' BAW filters and constitutes false advertising in violation of 15 U.S.C. § 1125(a).

189. Akoustis' description of its BAW filters as being "single crystal" is made in commercial advertising or promotion, including via Akoustis' website, presentations, and press releases. Akoustis also describes its BAW filters as being "single crystal" in its SEC filings.

190. Akoustis falsely or misleadingly represents to the consuming public, customers, potential customers (collectively, the "target customers") and investors that Akoustis' BAW filters

are of a “single crystal” construction to support a claim of superior performance over the competition. However, Akoustis’ BAW filters are, in fact, polycrystalline.

191. Akoustis’ false or misleading representations that its BAW filters are “single crystal” actually deceives a substantial portion of the target customers, or has the tendency to deceive the target customers.

192. On information and belief, Akoustis used its false or misleading representations of its BAW filters as “single-crystal” to promote and advertise that it has better technology than Qorvo, or that its BAW filters have better performance than Qorvo’s filters. Akoustis’ false or misleading representations that its BAW filters are “single crystal” are material because they are likely to influence the purchasing decisions of the target customers.

193. Akoustis’ false or misleading representations that its BAW filters are “single crystal” involves BAW filters that are advertised, promoted, sold, and distributed in interstate commerce, including via Akoustis’ website.

194. Akoustis’ false or misleading representations that its BAW filters are “single crystal” have competitively injured, and are likely to further competitively injure, Qorvo by diverting sales from target customers and through loss of their goodwill.

195. Akoustis knows that its representations of its BAW filters as being of a “single crystal” construction are false or misleading.

196. Akoustis’ false or misleading representations of fact are done with bad faith and malice or reckless indifference to Qorvo’s and consumers’ interests.

197. Based upon the above wrongful acts of Akoustis, Qorvo has incurred monetary damages through the diversion of sales and loss of goodwill in an amount to be fully demonstrated at trial.

198. Akoustis' intentional and bad faith or misleading representations of fact will continue until enjoined by this Court.

199. Qorvo is entitled to preliminary and permanent injunctive to prevent Defendant's continued false advertising.

D. Count IV – False Patent Marking Under 35 U.S.C. § 292

200. Qorvo incorporates by reference the allegations of paragraphs 1 through 199 set forth above.

201. Akoustis has marked each of the Accused Products with U.S. Patent No. 10,256,786 on datasheets available on its website (*see* ALL PRODUCTS – AKOUSTIS TECHNOLOGIES, <https://akoustis.com/products/filters/applications/all-products/> (last visited Sept. 14, 2021)).

202. The independent claims of U.S. Patent No. 10,256,786 recite piezoelectric layers. Based on the disclosure of U.S. Patent No. 10,256,786 and Akoustis' statements, the recited piezoelectric layers are single-crystal.

203. However, Akoustis' BAW filter products are not "single-crystal." They are polycrystalline. On information and belief, Akoustis regularly tests and analyzes its BAW filter products and knew or should have known that Akoustis' BAW filter products are polycrystalline. Moreover, Akoustis knew or should have known that U.S. Patent No. 10,256,786 covered only single-crystal piezoelectric layers.

204. Akoustis falsely marked the Accused Products with purpose or intent to deceive the public that the Accused Products were covered by U.S. Patent No. 10,256,786 in violation of 35 U.S.C. § 292. Akoustis engaged in the false marketing of its products to support a claim that the products are innovative and result in superior performance. Akoustis' false marking is material because it is likely to influence the purchasing decisions of the target customers. Akoustis' false

marking has competitively injured, and is likely to further competitively injure, Qorvo by diverting sales from target customers and through loss of their goodwill.

205. Qorvo is entitled to compensatory damages in an amount to be determined at trial.

E. Count V - Unfair and Deceptive Trade Practices Prohibited by NC. Gen. Stat. § 75-1

206. Qorvo incorporates by reference the allegations of paragraphs 1 through 205 set forth above.

207. By virtue of the aforesaid acts, Akoustis has engaged in unfair or deceptive acts or practices and in unfair methods of competition affecting commerce in the state of North Carolina. Such acts violate the North Carolina Unfair and Deceptive Trade Practices Act, NC. Gen. Stat. § 75-1.1.

208. Qorvo owns and possesses the Qorvo BAW Proprietary Information and employed the Ex-Qorvo BAW Employees.

209. Qorvo's business is dependent on its ability to attract and keep key technical personnel and management. Qorvo has invested substantial time, effort, and expense in developing its employees' knowledge and experience in the areas crucial to Qorvo's BAW filter technologies, including in RF engineering, integrated circuit and filter design, and technical marketing and support.

210. Akoustis has engaged in conscious, methodical, and systematic recruitment and solicitation of Qorvo's employees in all major areas. On information and belief, a large number of Akoustis' current employees are Ex-Qorvo employees. As detailed above, these include many employees with significant technical and business experience crucial to BAW filter technology, such as the Ex-Qorvo BAW Employees.

211. Akoustis is seeking to damage Qorvo's business through the recruiting and hiring away from Qorvo the Ex-Qorvo BAW Employees. This poaching is intended to damage Qorvo by attrition, rather than by fair competition in the marketplace.

212. Moreover, as described above, the Ex-Qorvo BAW Employees have significant knowledge of the Qorvo BAW Proprietary Information, and were placed by Akoustis in positions with substantial responsibility that overlapped with those employees' previous roles at Qorvo such that disclosure of the Qorvo BAW Proprietary Information was inevitable. On information and belief, Akoustis targeted the Ex-Qorvo BAW Employees, including Robert Dry, David Breton, Guillermo Moreno, William Schmid, Kindra Lane, Wendy Wright, Paul Makowskyj, Tony Espinoza, Ali-Imran Bawangaonwala, Todd Bender, Shawn Gibb, David Hodge, Rohan Houlden, Joonbum Kwon, John Myrick, and Ya Shen to obtain Qorvo BAW Proprietary Information from those employees. On information and belief, the goal of the systematic raiding of Qorvo employees was to use this Qorvo BAW Proprietary information as a short-cut for Akoustis to enter the market for BAW filters.

213. The Qorvo BAW Proprietary Information derives independent economic value, actual or potential, from not being known generally to, and not being readily ascertainable through proper means by, another person who can obtain economic value from the disclosure or use of the information.

214. Qorvo expended substantial effort, time, and capital in developing, maintaining, and possessing the Qorvo BAW Proprietary Information that it incorporated into its highly successful BAW filter products.

215. During their employment by Qorvo, the Ex-Qorvo BAW Employees hired by Akoustis had access to the Qorvo BAW Proprietary Information.

216. Following the termination of their employment, the Ex-Qorvo BAW Employees were not authorized to disclose, share, use, or possess the Qorvo BAW Proprietary Information.

217. Akoustis was aware that the Ex-Qorvo BAW Employees it recruited and hired had entered into binding agreements with Qorvo that prohibited the employees from taking, disclosing, using, sharing, or possessing the Qorvo BAW Proprietary Information.

218. On information and belief, Akoustis requested the Ex-Qorvo BAW Employees to disclose, use, or share the Qorvo BAW Proprietary Information as part of Akoustis' recruitment process, or otherwise expected the Ex-Qorvo BAW Employees to disclose, use, or share, the Qorvo BAW Proprietary Information as part of their employment at Akoustis, or implemented a hiring scheme that resulted in the Ex-Qorvo BAW Employees inevitably disclosing, using, or sharing, the Qorvo BAW Proprietary Information as part of their employment at Akoustis.

219. As set forth above, one example of Akoustis' efforts to obtain Qorvo BAW Proprietary Information as part of the recruitment process occurred in late 2020 when an Akoustis engineering manager asked a Qorvo RF Product Engineer to access Qorvo's computer systems during a job interview and provide screen shots of Qorvo BAW Proprietary information to Akoustis.

220. On information and belief, Ex-Qorvo BAW Employees have disclosed, shared, or used the Qorvo BAW Proprietary Information, at least inevitably, as part of their job responsibilities at Akoustis.

221. On information and belief, the Ex-Qorvo BAW Employees will continue to disclose, share, or use the Qorvo BAW Proprietary Information as part of their job responsibilities at Akoustis unless and until they are enjoined by this Court from doing so.

222. On information and belief, the poaching of the Ex-Qorvo BAW Employees and the misappropriation of the Qorvo BAW Proprietary Information has resulted in Akoustis being able to bring its “3-7 GHz BAW RF filter portfolio,” including the Accused Products, to market in a timeframe that is highly unlikely, if not impossible, without these unfair practices. Akoustis’ “3-7 GHz BAW RF filter portfolio,” has been placed in commerce in North Carolina, and interstate and foreign commerce, within the past four years.

223. As a result of Akoustis’ wrongful conduct, Qorvo has been substantially and irreparably harmed in an amount not readily capable of determination and for which there is no adequate remedy at law. Unless restrained by this Court, Akoustis will cause further irreparable injury to Plaintiff.

224. Qorvo is entitled to preliminary and permanent injunctive relief enjoining Akoustis, its agents and employees, and all persons acting in concert or participation with Akoustis, from engaging in any further poaching of Qorvo employees and/or use of the Qorvo BAW Proprietary Information.

225. Moreover, Akoustis has engaged in unfair or deceptive acts or practices and unfair methods of competition affecting commerce in the state of North Carolina through Akoustis’ false or misleading representations to the target customers and investors that Akoustis’ BAW filters are of a “single crystal” construction. Akoustis’ false or misleading representations that its BAW filters are “single crystal” actually deceives, or has the tendency to deceive, the target customers, and its representations are material because they are likely to influence the purchasing decisions of the target customers. Akoustis’ false or misleading representations have competitively injured, and are likely to further competitively injure, Qorvo by diverting sales from target customers and through loss of their goodwill.

226. On information and belief, Akoustis has made and will continue to make substantial profits and/or gains to which it is not in law or equity entitled.

227. On information and belief, Akoustis' violations of NC. Gen. Stat. § 75-1.1 have directly and proximately caused and continue to cause Qorvo substantial actual damages in North Carolina and throughout the United States, and Qorvo is entitled to recover actual damages in an amount to be proven at trial.

228. Pursuant to NC. Gen. Stat. § 75-16 and NC. Gen. Stat. § 75-16.1 Qorvo is entitled to treble damages and attorney fees for Akoustis' unfair and deceptive trade practices and unfair methods of competition.

F. Count VI – Misappropriation of Trade Secrets Under the Defense of Trade Secrets Act (18 U.S.C. § 1832 *et seq.*)

229. Qorvo incorporates by reference the allegations of paragraphs 1 through 228 set forth above.

230. By virtue of the aforesaid acts, which upon information and belief, are continuing, Akoustis has and continues to engage in trade secret misappropriation under federal law. Such acts violate the Defense of Trade Secrets Act (18 U.S.C. § 1832 *et seq.*).

231. As set forth in paragraphs 61 through 153, Qorvo has developed an extensive trade secret portfolio. Qorvo's trade secrets relate to, among other things, documentation and know-how for designing, developing, modeling, simulating, testing, qualifying, manufacturing, marketing, selling, and supporting its BAW filters and other products, as well as Qorvo's compensation and job structure for its engineers and other employees, including those who work on Qorvo's BAW filters and other products. Use of Qorvo's trade secrets allowed Akoustis to implement technology that would otherwise take months if not years to develop independently and to avoid dead-ends.

[REDACTED]

In the end, Akoustis was able to get to market much faster compared to a company that did not have the benefit of Qorvo's trade secrets.

232. As set forth in paragraphs 1 through 29, as well as in paragraphs 61 through 153, Qorvo has taken and continues to take commercially reasonable steps to protect its trade secret information. Qorvo protects its trade secrets at least in part by maintaining the confidentiality of documents containing such information.

233. As set forth in paragraphs 1 through 29, as well as in paragraphs 61 through 153, Qorvo's trade secrets derive independent actual or potential commercial value from not being generally known or readily ascertainable through independent development or reverse engineering by persons who can obtain value from the trade secrets' disclosure. Maintaining the confidentiality of Qorvo's trade secrets gives Qorvo a highly valuable advantage over competitors in the BAW filter market.

234. Qorvo's trade secrets and their use by Akoustis implicate interstate or foreign commerce. For example, both Qorvo and Akoustis sell BAW filters in interstate and foreign commerce and the trade secrets were used by both parties to develop such filters.

235. As set forth in paragraphs 1 through 29, as well as in paragraphs 61 through 153, Akoustis improperly acquired, disclosed, used, appropriated, took, carried away, concealed, copied, duplicated, downloaded, replicated, transmitted, sent, uploaded, communicated, or conveyed the Qorvo trade secrets for the benefit of Akoustis.

236. As set forth in paragraphs 1 through 29, as well as in paragraphs 61 through 153, Akoustis' improper misappropriations were facilitated through former Qorvo employees who Akoustis recruited for the purpose of obtaining such trade secrets and confidential information, and who—during the time period prior to their departure from Qorvo and their joining Akoustis—

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had a specific opportunity to improperly access and take documents containing Qorvo BAW Proprietary Information and other Qorvo confidential information, including for disclosure or use at Akoustis without the express or implied consent or authority Qorvo. Furthermore, at least because Akoustis recruited and hired these employees (those who had accessed and taken such Qorvo documents), Akoustis, as their new employer, had a specific opportunity to improperly obtain and use the documents, and the trade secrets within them, without the express or implied consent or authority of Qorvo.

237. The use of Qorvo's trade secrets by Akoustis was without Qorvo's authorization. Qorvo did not consent to their acquisition, disclosure, or use of Qorvo's trade secrets.

238. Akoustis intended to, and continues to, convert Qorvo's trade secrets to the economic benefit of one other than their owner, Qorvo.

239. Akoustis knew and intended that Qorvo, as the owner of Qorvo's trade secrets, would be injured by their actions.

240. As a result of Akoustis' misappropriation of Qorvo's trade secrets, Qorvo has suffered actual damages in an amount to be proven at trial.

241. As a result of Akoustis' misappropriation, Akoustis has been unjustly enriched.

242. Qorvo further pleads entitlement to a reasonable royalty to compensate Qorvo for Akoustis' misappropriation of trade secrets.

243. Qorvo is informed and believes, and thereon alleges, that Akoustis' misappropriation of Qorvo's trade secrets was willful and malicious based on the facts alleged herein. Akoustis acted with a purpose and willingness to commit the acts alleged, and Akoustis' conduct was not reasonable under the circumstances. Qorvo is therefore entitled to exemplary damages and attorney fees and costs. Qorvo further seeks exemplary damages against Akoustis in

an amount up to two times the amount of Qorvo's actual damages according to proof under 18 U.S.C. § 1836.

244. The misappropriation of Qorvo's trade secrets has caused and will continue to cause Qorvo irreparable and substantial injury and therefore cannot be fully redressed through damages alone.

245. If Akoustis were permitted to continue to use and disseminate Qorvo's trade secrets, Qorvo will be irreparably harmed and the economic damages to Qorvo will be difficult to quantify. An injunction prohibiting Akoustis from further acquisition, disclosure, use, and possession of Qorvo's trade secrets is necessary to provide Qorvo with complete relief.

246. On information and belief, Akoustis has made and will continue to make substantial profits and/or gains on account of its misappropriations, to which it is not in law or equity entitled.

247. On information and belief, Akoustis' violations of 18 U.S.C. § 1832 *et seq.* have directly and proximately caused and continue to cause Qorvo substantial actual damages, and Qorvo is entitled to recover actual damages in an amount to be proven at trial.

**G. Count VII – Racketeer Influenced and Corrupt Organizations Act
(18 U.S.C. § 1961 *et seq.*)**

248. Qorvo incorporates by reference the allegations of paragraphs 1 through 247 set forth above.

249. Akoustis is an enterprise because it is a Delaware corporation.

250. As set forth in paragraphs 61 through 153, Akoustis conspired with Rohan Houlden, Robert Dry, and other former Qorvo employees, before Akoustis formally hired such employees at a later date, to steal, copy, possess and use Qorvo's trade secrets without authorization, violating the Defense of Trade Secrets Act. Akoustis' prior and continuing conspiracy to misappropriate

Qorvo's trade secrets through improper means constitutes unlawful activity pursuant to 18 U.S.C. § 1962(d).

251. As set forth in paragraphs 61 through 153, Akoustis has misappropriated Qorvo's trade secrets in violation of the Defense of Trade Secrets Act. On multiple occasions, persons employed by Akoustis—e.g., Rohan Houlden, Robert Dry—have stolen Qorvo's trade secrets, have shared Qorvo's trade secrets, and have re-purposed Qorvo's trade secrets for Akoustis' use to gain an unfair competitive advantage in the market for BAW filters. Akoustis' multiple and continuing violations of the Defense of Trade Secrets Act and misappropriation through improper means of Qorvo's trade secrets constitutes pattern of racketeering activity pursuant to 18 U.S.C. § 1962(c).

252. Akoustis' racketeering activities is ongoing because, on belief, Akoustis continues to use Qorvo's trade secrets in its business including, but not limited to, documentation and know-how for designing, developing, modeling, simulating, testing, qualifying, manufacturing, marketing, selling, and supporting its BAW filters and other products, as well as Qorvo's compensation and job structure for its engineers and other employees, including those who work on Qorvo's BAW filters and other products.

253. Akoustis' racketeering activities are related to Akoustis and the former Qorvo employees using misappropriated Qorvo trade secrets in interstate and foreign commerce by, among other things, using know-how and documentation for designing, developing, modeling, simulating, testing, qualifying, manufacturing, marketing, selling, and supporting Akoustis BAW filters and other products, as well as Qorvo's compensation and job structure for its engineers and other employees.

254. On information and belief, Akoustis' violations of 18 U.S.C. § 1961 *et seq.* have directly and proximately caused and continue to cause Qorvo substantial actual damages, and Qorvo is entitled to recover actual damages in an amount to be proven at trial.

255. Pursuant to 18 U.S.C. § 1964(c), Qorvo is entitled to treble damages, costs, and attorney fees for Akoustis' racketeering.

H. Count VIII – Misappropriation of Trade Secrets Under the North Carolina Trade Secrets Protection Act (N.C. Gen. Stat § 66-152 *et seq.*)

256. Qorvo incorporates by reference the allegations of paragraphs 1 through 255 set forth above.

257. By virtue of Akoustis' acts of misappropriation, including as set forth in paragraphs 61 through 153, which upon information and belief are continuing, Akoustis has and continues to engage in trade secret misappropriation under North Carolina state law. Such acts violate the North Carolina Trade Secrets Protection Act (N.C. Gen. Stat § 66-152 *et seq.*).

258. As set forth in paragraphs 61 through 153, Qorvo has developed an extensive trade secret portfolio. Qorvo's trade secrets relate to, among other things, documentation and know-how for designing, developing, modeling, simulating, testing, qualifying, manufacturing, marketing, selling, and supporting its BAW filters and other products, as well as Qorvo's compensation and job structure for its engineers and other employees, including those who work on Qorvo's BAW filters and other products. Use of Qorvo's trade secrets allowed Akoustis to implement technology that would otherwise take months if not years to develop independently and to avoid dead-ends. In the end, Akoustis was able to get to market much faster compared to a company that did not have the benefit of Qorvo's trade secrets.

259. As set forth in paragraphs 1 through 29, as well as in paragraphs 61 through 153, Qorvo has taken and continues to take commercially reasonable steps to protect its trade secret

information. Qorvo protects its trade secrets at least in part by maintaining the confidentiality of documents containing such information.

260. As set forth in paragraphs 1 through 29, as well as in paragraphs 61 through 153, Qorvo's trade secrets derive independent actual or potential commercial value from not being generally known or readily ascertainable through independent development or reverse engineering by persons who can obtain value from the trade secrets' disclosure. Maintaining the confidentiality of Qorvo's trade secrets gives Qorvo a highly valuable advantage over competitors in the BAW filter market.

261. Qorvo's trade secrets and their use by Akoustis implicate interstate or foreign commerce. For example, both Qorvo and Akoustis sell BAW filters in interstate and foreign commerce and the trade secrets were used by both parties to develop such filters.

262. As set forth in paragraphs 1 through 29, as well as in paragraphs 61 through 153, Akoustis improperly acquired, disclosed, used, appropriated, took, carried away, concealed, copied, duplicated, downloaded, replicated, transmitted, sent, uploaded, communicated, or conveyed the Qorvo trade secrets for the benefit of Akoustis.

263. As set forth in paragraphs 1 through 29, as well as in paragraphs 61 through 153, Akoustis' improper misappropriations were facilitated through former Qorvo employees who Akoustis recruited for the purpose of obtaining such trade secrets and confidential information, and who—during the time period prior to their departure from Qorvo and their joining Akoustis—had a specific opportunity to improperly access and take documents containing Qorvo BAW Proprietary Information and other Qorvo confidential information, including for disclosure or use at Akoustis without the express or implied consent or authority Qorvo. Furthermore, at least because Akoustis recruited and hired these employees (those who had accessed and taken such

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Qorvo documents), Akoustis, as their new employer, had a specific opportunity to improperly obtain and use the documents, and the trade secrets within them, without the express or implied consent or authority of Qorvo.

264. The use of Qorvo's trade secrets by Akoustis was without Qorvo's authorization. Qorvo did not consent to their acquisition, disclosure, or use of Qorvo's trade secrets.

265. Akoustis intended to, and continues to, convert Qorvo's trade secrets to the economic benefit of one other than their owner, Qorvo.

266. Akoustis knew and intended that Qorvo, as the owner of Qorvo's trade secrets, would be injured by their actions.

267. As a result of Akoustis' misappropriation of Qorvo's trade secrets, Qorvo has suffered actual damages in an amount to be proven at trial.

268. As a result of Akoustis' misappropriation, Akoustis has been unjustly enriched.

269. Qorvo further pleads entitlement to a reasonable royalty to compensate Qorvo for Akoustis' misappropriation of Qorvo's trade secrets.

270. Qorvo is informed and believes, and thereon alleges, that Akoustis' misappropriation of Qorvo's trade secrets was willful and malicious based on the facts alleged herein. Akoustis acted with a purpose and willingness to commit the acts alleged, and Akoustis' conduct was not reasonable under the circumstances. Qorvo is therefore entitled to exemplary damages and attorney fees and costs.

271. The misappropriation of Qorvo's trade secrets has caused and will continue to cause Qorvo irreparable and substantial injury and therefore cannot be fully redressed through damages alone.

272. If Akoustis were permitted to continue to use and disseminate Qorvo's trade secrets, Qorvo will be irreparably harmed and the economic damages to Qorvo will be difficult to quantify. An injunction prohibiting Akoustis from further acquisition, disclosure, use, and possession of Qorvo's trade secrets is necessary to provide Qorvo with complete relief.

I. Count IX – Civil Conspiracy Under North Carolina

273. Qorvo incorporates by reference the allegations of paragraphs 1 through 272 set forth above.

274. By virtue of Akoustis' acts of misappropriation, including as set forth in paragraphs 6161 through 153, Akoustis and its employees engaged in trade secret misappropriation in violation of the North Carolina Trade Secrets Protection Act (N.C. Gen. Stat § 66-152 *et seq.*).

275. Upon information and belief, Akoustis conspired with one or more Qorvo employees, including former Qorvo employees who later joined Akoustis (but prior to their employment by Akoustis), to violate their legal obligations to Qorvo and misappropriate Qorvo's trade secrets for Akoustis' benefit. Upon information and belief, it was agreed that if these employees would join Akoustis and aid Akoustis in this manner, Akoustis' resultant success in the marketplace would yield substantial financial increase for these employees including through the increased value of securities in Akoustis that would be granted to such employees pursuant to their anticipated employment with Akoustis. Upon information and belief, Akoustis and the one or more employees committed acts in furtherance of the aforesaid conspiracy.

276. Akoustis' prior and continuing conspiracy with such employees to misappropriate Qorvo's trade secrets and confidential information is causing, and threatens to continue causing, Qorvo to suffer irreparable harm, including but not limited to loss of business advantage, loss of exclusivity rights, and loss of its investment in its trade secrets.

277. As a result of Akoustis' acts of conspiracy in connection with trade secret misappropriation, Qorvo has suffered actual damages in an amount to be proven at trial, and Akoustis has been unjustly enriched.

VI. PRAYER FOR RELIEF

WHEREFORE, Plaintiff Qorvo respectfully prays for relief as follows:

- 1) Enter a judgment that Akoustis has directly and indirectly infringed the Patents-in-Suit;
- 2) Award Qorvo damages in an amount sufficient to compensate it for Akoustis' infringement of the Patents-in-Suit, together with pre-judgment and post-judgment interest and costs, and all other damages permitted under 35 U. S. C. § 284;
- 3) Find that Akoustis' infringement of the Patents-in-Suit has been willful and deliberate;
- 4) Enter a judgment awarding Qorvo treble damages as a result of Akoustis' willful and deliberate infringement of the Patents-in-Suit, pursuant to 35 U.S.C. § 284;
- 5) Find that Akoustis' actions render this an exceptional case under 35 U.S.C. § 285 and award Qorvo its attorneys' fees, costs and expenses;
- 6) Preliminarily and permanently enjoin, pursuant to 35 U.S.C. § 283, Akoustis, and all affiliates, employees, agents, officers, directors, attorneys, successors, and assigns, and all those acting on behalf of or in active concert or participation with any of them, from further infringing the Patents-in-Suit;
- 7) Preliminarily and permanently enjoin Akoustis, and all affiliates, employees, agents, officers, directors, attorneys, successors, and assigns, and all those acting on behalf of or in active concert or participation with any of them, from further falsely advertising the properties of its BAW filters;

8) Award Qorvo damages in an amount sufficient to compensate it for Akoustis' false advertising;

9) Preliminarily and permanently enjoin Akoustis, and all affiliates, employees, agents, officers, directors, attorneys, successors, and assigns, and all those acting on behalf of or in active concert or participation with any of them, from further falsely marking its BAW filters;

10) Award Qorvo damages in an amount sufficient to compensate it for Akoustis' false patent marking;

11) Award Qorvo damages in an amount sufficient to compensate it for Akoustis' trade secret misappropriation under the North Carolina Trade Secrets Protection Act (N.C. Gen. Stat § 66-152 *et seq.*) and/or under the Defense of Trade Secrets Act (18 U.S.C. § 1832 *et seq.*);

12) Award Qorvo damages, including exemplary damages, punitive damages, attorneys' fees, and costs and expenses in view of Akoustis' violations under the North Carolina Trade Secrets Protection Act (N.C. Gen. Stat § 66-152 *et seq.*) and/or under the Defense of Trade Secrets Act (18 U.S.C. § 1832 *et seq.*);

13) Preliminarily and permanently enjoin Akoustis, and all affiliates, employees, agents, officers, directors, attorneys, successors, and assigns, and all those acting on behalf of or in active concert or participation with any of them, from further misappropriating and using Qorvo's trade secrets under the North Carolina Trade Secrets Protection Act (N.C. Gen. Stat § 66-152 *et seq.*);

14) Preliminarily and permanently enjoin Akoustis, and all affiliates, employees, agents, officers, directors, attorneys, successors, and assigns, and all those acting on behalf of or in active concert or participation with any of them, from further misappropriating and using Qorvo's trade secrets under the Defense of Trade Secrets Act (18 U.S.C. § 1832 *et seq.*);

15) Preliminarily and permanently enjoin Akoustis, and all affiliates, employees, agents, officers, directors, attorneys, successors, and assigns, and all those acting on behalf of or in active concert or participation with any of them, from further selling any products that resulted, in whole or in part, from misappropriating Qorvo's trade secrets under the North Carolina Trade Secrets Protection Act (N.C. Gen. Stat § 66-152 *et seq.*);

16) Preliminarily and permanently enjoin Akoustis, and all affiliates, employees, agents, officers, directors, attorneys, successors, and assigns, and all those acting on behalf of or in active concert or participation with any of them, from further misappropriating and using Qorvo's trade secrets under the Defense of Trade Secrets Act (18 U.S.C. § 1832 *et seq.*);

17) Award Qorvo damages, including exemplary damages, punitive damages, attorneys' fees, and costs and expenses in view of Akoustis' wrongful acts committed pursuant to a conspiracy in violation of North Carolina law;

18) Award Qorvo damages in an amount sufficient to compensate it for Akoustis' violation under 18 U.S.C. § 1961 *et seq.*;

19) Award Qorvo damages, including treble damages, punitive damages, attorneys' fees, and costs and expenses in view of Akoustis' violations under 18 U.S.C. § 1961 *et seq.*;

20) Preliminarily and permanently enjoin Akoustis, and all affiliates, employees, agents, officers, directors, attorneys, successors, and assigns, and all those acting on behalf of or in active concert or participation with any of them, from further selling any products that resulted, in whole or in part, from pattern of racketeering activity under 18 U.S.C. § 1961 *et seq.*;

21) Preliminarily and permanently enjoin Akoustis, and all affiliates, employees, agents, officers, directors, attorneys, successors, and assigns, and all those acting on behalf of or

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in active concert or participation with any of them, from further misappropriating and using Qorvo's trade secrets under 18 U.S.C. § 1961 *et seq.*;

22) Award Qorvo damages in an amount sufficient to compensate it for Akoustis' unfair and deceptive trade practices;

23) Award Qorvo damages, including treble damages, punitive damages, and attorneys' fees, in view of Akoustis' violation of N.C. Gen. Stat. § 75-1.1;

24) Preliminarily and permanently enjoin Akoustis, and all affiliates, employees, agents, officers, directors, attorneys, successors, and assigns, and all those acting on behalf of or in active concert or participation with any of them, from further unfair and deceptive trade practices pursuant to N.C. Gen. Stat. § 75-1.1;

25) Order a full and complete accounting to Qorvo for Akoustis' profits, gains, advantages or the value of business opportunities received from the foregoing acts of infringement;

26) Award Qorvo all actual, compensatory, consequential, liquidated, treble, and special damages to which it is entitled in an amount to be determined at trial;

27) Assess punitive damages against Akoustis;

28) Award Qorvo all attorneys' fees and costs to which it is entitled associated with bringing and prosecuting this action;

29) Award pre-judgment and post-judgment interest against Akoustis; and

30) Award Qorvo such other and further relief as the Court deems just and proper.

DEMAND FOR JURY TRIAL

Pursuant to Rule 38(b) of the Federal Rules of Civil Procedure and D. Del. LR 38.1, Qorvo demands a trial by jury on all issues raised by the Complaint.

Dated: February 8, 2023

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